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A transcriptional biosensor reveals mechanisms of α-ketoglutarate signaling to chromatin

Alex C. Sternisha†, Haocheng Li†, *et al.*

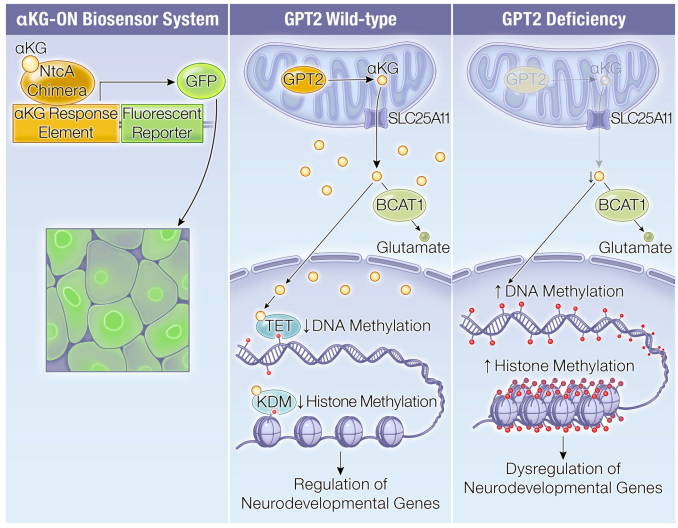
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INTRODUCTION: The metabolite α-ketoglutarate (αKG) is a substrate for dioxygenase enzymes that demethylate histones and DNA in the nucleus, influencing gene expression, cell fate, and tumor suppression. Despite the importance of nuclear αKG for chromatin-modifying enzyme function, mechanisms that govern this metabolite pool remain poorly defined. Revealing these mechanisms is vital for understanding how appropriate chromatin structure and gene expression patterns are maintained under homeostatic conditions and dysregulated in disease.

RATIONALE: Deciphering how nuclear αKG abundance is regulated in human cells has been challenging for several reasons. First, methods developed to study compartmentalized metabolism in mitochondria, lysosomes, and other organelles are not readily transferrable to nuclei. Second, intracellular αKG is regulated by a complex network of more than 100 enzymes and transporters that act directly on this metabolite. To address these challenges, we sought to create a biosensor capable of detecting changes in nuclear αKG and use it to identify the molecular processes that govern this metabolite pool.

RESULTS: We engineered the transcriptional "αKG-ON biosensor system" by leveraging NtcA, an αKG-responsive cyanobacterial transcription factor. Chimeric NtcA proteins were expressed together with a fluorescent reporter gene driven by a synthetic promoter featuring NtcA binding sites derived from cyanobacterial genomes. Together, these elements enabled fluorescence-based monitoring of the nuclear αKG pool in living human cells. We used the αKG-ON biosensor system in a forward genetic screen to identify genes that play dominant roles in controlling αKG abundance in nuclei. This approach uncovered an interorganelle pathway that mediates the transfer of αKG from mitochondria to nuclei. Within this pathway, mitochondrial glutamic-pyruvic transaminase 2 (GPT2) synthesizes αKG that is preferentially exported to the cytosol through the SLC25A11 transporter, a component of the malate-aspartate shuttle. Once in the cytosol, αKG may enter the nucleus or undergo catabolism to glutamate by the branched-chain amino acid transaminase 1 (BCAT1). Our findings align with prior research showing that BCAT1 competes with chromatin demethylating enzymes for access to nucleocytosolic αKG and reveal sequential activities of GPT2 and SLC25A11 as key suppliers of this metabolite pool.

Disrupting this interorganelle pathway of αKG metabolism caused chromatin hypermethylation in cultured cells in vitro and in tissues in vivo. A mouse model of GPT2 deficiency, an inborn error of metabolism caused by inactivating mutations in *GPT2*, displayed increases in methylation of both DNA and histones in brain tissue coupled with profound dysregulation of neurodevelopmental gene expression programs. Administering an esterified form of αKG to Gpt2-null mice restored chromatin methylation patterns, normalized gene expression, and improved mouse fitness, establishing a causal link between nuclear αKG depletion and the molecular pathogenesis of GPT2 deficiency.



Mechanisms governing αKG-dependent chromatin demethylation. Dioxygenases that demethylate DNA and histones require αKG. The αKG-ON biosensor system enables monitoring of the nuclear αKG pool that sustains activity of these enzymes. The mitochondrial GPT2 transaminase and SLC25A11 transporter act together to supply αKG to the nucleocytosolic compartment. In GPT2 deficiency, this pathway is impaired, causing chromatin hypermethylation and dysregulation of neurodevelopmental genes. GFP, green fluorescent protein; TET, ten-eleven translocation methylcytosine dioxygenases; KDM, histone lysine demethylase. [Figure by Melissa Logies]

CONCLUSION: This study provides a tool for monitoring αKG abundance in the nuclei of human cells and defines a pathway for mitochondrion-to-nucleus transfer of αKG that sustains epigenomic regulation. These findings underscore compartmentalized metabolism as a critical determinant of epigenetic state and reveal nuclear αKG depletion as a driver of neurodevelopmental defects in GPT2 deficiency. Moreover, our work suggests that αKG supplementation represents a potential strategy to counteract pathogenic mechanisms in patients with this disorder. □

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CELL BIOLOGY

A transcriptional biosensor reveals mechanisms of α-ketoglutarate signaling to chromatin

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The metabolite α-ketoglutarate (αKG) is required for chromatin demethylation, but mechanisms that control αKG abundance in the nucleus are poorly defined. We designed a biosensor to monitor this metabolite pool in human cells using an αKG-responsive cyanobacterial transcription factor, NtcA, and used it to identify genes that regulate αKG in the nucleus. We defined an interorganelle pathway in which sequential mitochondrial activities of glutamic-pyruvic transaminase 2 (GPT2) and the SLC25A11 transporter supply nuclear αKG. In a mouse model of GPT2 deficiency, an inborn error of metabolism, Gpt2 loss caused histone hypermethylation in the brain and dysregulated neurodevelopmental genes. Restoring αKG counteracted these changes and promoted mouse fitness. Our work provides a tool to directly monitor nuclear αKG and reveals nuclear αKG depletion as a key pathogenic mechanism underlying GPT2 deficiency.

Metabolites serve as substrates for enzymes that chemically modify DNA and histones (*1*, *2*). Metabolites such as acetyl–coenzyme A (acetyl-CoA) and oxidized nicotinamide adenine dinucleotide (NAD^+) regulate chromatin acetylation, whereas S-adenosylmethionine (SAM) and α-ketoglutarate (αKG) regulate methylation (*3*, *4*). αKG plays a key role in cell fate control and tumor suppression because dysregulation of αKG availability alters cell-state transitions (*5*–*9*). Recurrent, cancer-associated mutations in metabolic enzymes produce oncometabolites that directly interfere with αKG signaling to chromatin and prime cells for malignant transformation (*10*–*18*). Despite this important regulatory role for the nuclear αKG pool, we have limited insights into the molecular mechanisms that control it. This is due to a paucity of experimental tools to measure nucleus-specific αKG in mammalian cells, as well as the numerous enzymes and transporters that act on αKG.

We pursued a synthetic biology approach to monitor the nuclear αKG pool in living human cells, motivated by an αKG-sensing mechanism in cyanobacteria. Cyanobacteria assimilate nitrogen by scavenging ammonium and using it to synthesize glutamine and glutamate from αKG (*19*). Under nitrogen starvation, αKG accumulates and is sensed by the transcription factor NtcA (*20*). αKG binding increases the affinity of NtcA for DNA (*21*), enabling activation of genes involved in the response to nitrogen deprivation (*22*). We hypothesized that a chimeric NtcA transcription factor could be adapted to drive expression of a synthetic reporter gene that serves as a proxy for nuclear αKG pool size in human cells. This would allow for unbiased, forward genetic approaches to identify genes that regulate cellular αKG metabolism and play dominant roles in controlling abundance of this metabolite in the nucleus.

Designing and optimizing the αKG-ON biosensor system to monitor nuclear αKG

To address the paucity of tools to measure nuclear αKG in human cells, we sought to adapt a cyanobacterial, αKG-regulated transcription factor, NtcA (*23*), to drive expression of a reporter gene in a nuclear αKG–dependent manner. This biosensor (hereafter, the αKG-ON biosensor system) included a chimeric NtcA transcription factor and a promoter-reporter gene DNA element that can be transactivated by NtcA (Fig. 1A). We posited that fusing NtcA to a nuclear localization signal (NLS) and transactivation domain would allow us to direct NtcA chimeras to the nucleus and drive transcription in human cells. Furthermore, engineering a synthetic promoter with NtcA binding sites (hereafter, the αKG response element, or αKG-RE) and placing it upstream of a green fluorescent protein (GFP) reporter gene would allow us to monitor changes in nuclear αKG. This design schema is based on biochemical (*21*) and structural (*20*, *24*, *25*) studies establishing that αKG binding to NtcA homodimers triggers transcription factor binding to successive DNA major grooves (Fig. 1B). We generated purified, recombinant NtcA that specifically bound αKG with a dissociation constant (K_d) of 2.0 mM (fig. S1, A and B), which is in line with whole-cell αKG levels in human cells (*26*).

We first screened components of the αKG-ON biosensor system in human embryonic kidney (HEK) 293 cells (Fig. 1C and fig. S1C). To create αKG-RE promoters, we derived NtcA binding sites from the promoters of glutamine synthetase–encoding *glnA* or *glnN* genes in *Synechocystis* sp. PCC 6803 or *Pseudanabaena* sp. PCC 6903 genomes (*22*, *27*, *28*) (fig. S1, D and E). We incorporated binding sites either individually or in five-repeat tiles upstream of a minimal cytomegalovirus (CMV) promoter. To assess outputs of the system, we treated cells with or without cell-permeable dimethyl-αKG (dm-αKG), measured GFP expression with flow cytometry, and monitored the percentage of cells that shifted into a GFP^{high} population within each αKG-RE–expressing stable line. Several combinations of NtcA chimeras and αKG-RE promoters displayed basal GFP reporter transactivation (presumably prompted by endogenous nuclear αKG) that was enhanced by dm-αKG

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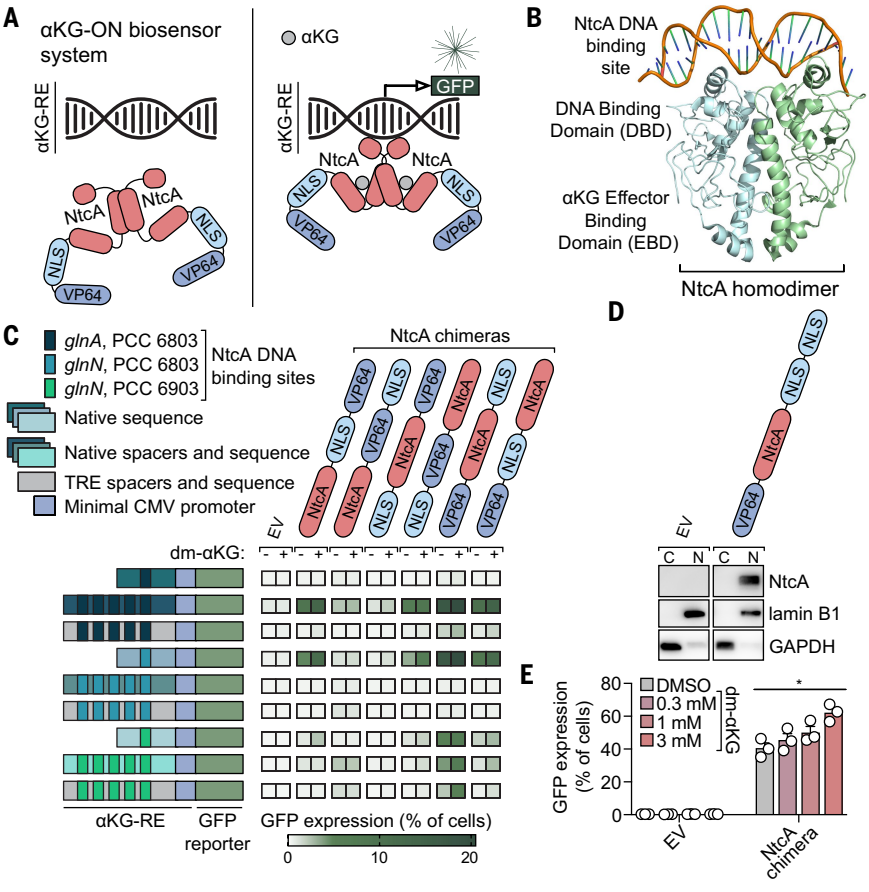


Fig. 1. Development and optimization of the αKG-ON biosensor system to monitor the nuclear αKG pool. (**A**) Schema of αKG-ON biosensor system transcriptional activity in the presence and absence of αKG. αKG-RE, αKG response element; NLS, nuclear localization signal; VP64, transcriptional activator Herpes Simplex Viral Protein 64; GFP, green fluorescent protein. (**B**) Cryo–electron microscopy structure of NtcA homodimer bound to DNA and αKG. Resolution was not sufficient to resolve αKG within this complex. Data are derived from Protein Data Bank accession 8H40 (*24*). (**C**) Pairwise combinatorial screen of αKG-ON biosensor system architectures in HEK293 cells. HEK293 lines stably expressing αKG-RE elements were transiently transfected with NtcA chimeras or an empty vector (EV; negative control). αKG-ON biosensor system activation (GFP expression) was quantified by means of flow cytometry in cells treated with 1 mM dimethyl-αKG (dm-αKG) or dimethyl sulfoxide (DMSO) for 48 hours before analysis. GFP expression (percent of cells) was determined by means of a gating strategy using EV-transfected, DMSO-treated cells as a baseline. (**D**) Immunoblot of the VP64-NtcA-2xNLS$_{SV40}$ chimera in nuclear (N; lamin B1 marker) and cytosolic [C; glyceraldehyde-3-phosphate dehydrogenase (GAPDH) marker] fractions prepared from HEK293 cells engineered to express the NtcA chimera or EV. (**E**) Flow cytometry quantification of αKG-ON biosensor system activation (GFP expression) in αKG-RE–expressing HEK293 cells transduced with the VP64-NtcA-2xNLS$_{SV40}$ chimera or EV and treated with DMSO or the indicated doses of dm-αKG for 48 hours. Data are means ± SEM. *$P < 0.05$ [ordinary one-way analysis of variance (ANOVA)].

supplementation. Notably, dm-αKG did not induce GFP in empty vector–expressing cells without NtcA chimeras. The αKG-RE promoter containing tiled NtcA binding sites from the PCC 6903 *glnN* promoter with tetracycline response element (TRE)–derived spacers was selected for further study.

We sought to optimize αKG-ON biosensor system performance by improving functionality of the NtcA chimeric transcription factor. Cell fractionation experiments showed that NtcA chimeras were not exclusively nuclear and that some chimera expression constructs produced truncated protein products (fig. S1F). In silico analyses identified cryptic nuclear export sequences in the VP64 peptide (fig. S1G). Eliminating VP16 monomer peptides that comprise VP64 enhanced nuclear localization but reduced biosensor output (fig. S1, H and I). Testing other mammalian transactivation domains (*29*–*32*) revealed that the Zta transactivation domain enhanced nuclear localization and eliminated truncated protein products (fig. S1J). We removed unnecessary methionine residues to prevent alternative translation products and replaced the c-myc–derived NLS with two SV40 large T-antigen–derived NLSs, which improved nuclear localization (fig. S1K). VP64-NtcA-2xNLS$_{SV40}$ displayed increased nuclear localization and αKG-RE promoter activation relative to other chimeras (Fig. 1, D and E, and fig. S1L). Furthermore, VP64-NtcA-2xNLS$_{SV40}$ exhibited enhanced or equivalent transcriptional activity compared with that of Zta-NtcA-NLS$_{c\text{-}myc}$ when expressed under promoters of varying strength (fig. S2, A to R). Thus, we selected the EF1α promoter–driven VP64-NtcA-2xNLS$_{SV40}$ chimera (or mutants thereof) for further studies.

The αKG-ON biosensor system responds to compartment-specific changes in αKG abundance

To assess specificity of the αKG-ON biosensor system, we produced a gradient of intracellular αKG levels in HEK293 cells by reducing glutamine (a source of αKG) or by supplementing dm-αKG (fig. S3A). We also created mutants of the VP64-NtcA-2xNLS$_{SV40}$ chimera that lack αKG (R90E, in which arginine at position 90 is replaced with glutamate) or DNA (ΔDBD) binding functions and expressed them or an empty vector in HEK293 cells. Only the wild-type (WT) NtcA chimera evoked a gradient of GFP expression that correlated with intracellular αKG (Fig. 2, A to H). Supplementing dm-αKG fully rescued decreased biosensor output caused by glutamine deprivation, indicating that this effect was αKG-dependent (Fig. 2I and fig. S4, A and B). WT and R90E NtcA chimeras (*20*, *33*) were used in tandem going forward and are hereafter referred to as "functional" and "inactive" versions of the αKG-ON biosensor system, respectively. We examined the effect of cell-cycle phase on biosensor output heterogeneity and observed increased GFP expression in both the functional and inactive arms during G_2/M phases, potentially because of increased reporter gene copy number (fig. S3, B and C). Biosensor output was not affected by supplementation with (*R*)-2-hydroxyglutarate-Bis(trifluoromethyl) benzidine [(*R*)-2HG-TFMB], a cell-permeable ester of the oncometabolite (*R*)-2HG (fig. S4, C to E) (*34*), or dimethyl succinate, a cell-permeable ester of the oncometabolite succinate (fig. S4, F to H). Expressing an IDH1-R132H cDNA led to (*R*)-2HG up-regulation (which was blocked by the mutant IDH1/IDH2 inhibitor vorasidenib) and histone hypermethylation (fig. S4, I and J). IDH1-mutant cells displayed a modest increase in αKG-ON biosensor system output (fig. S4K), but this effect was (*R*)-2HG–independent because it was not altered by vorasidenib. Given the structural similarity between (*R*)-2HG, succinate, and αKG, our findings indicate that biosensor output is uncoupled from processes of oncometabolite-driven chromatin hypermethylation. Thus, the αKG-ON biosensor system was specifically regulated by intracellular αKG levels and required both the αKG and DNA binding activities of the NtcA transcription factor.

αKG exchange across the inner mitochondrial membrane is controlled by dedicated transport mechanisms, partitioning mitochondrial αKG metabolism from that in other organelles. By contrast, the

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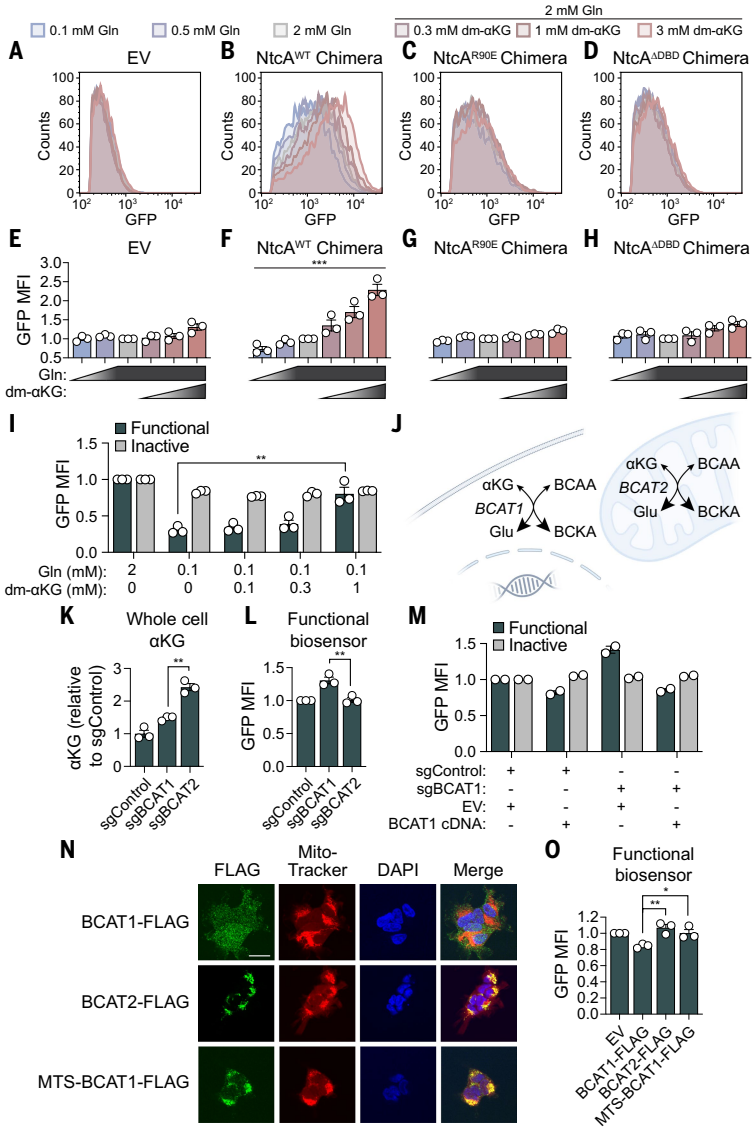


Fig. 2. The αKG-ON biosensor system responds to metabolic and genetic perturbations of the nuclear αKG pool. (**A** to **D**) Representative histograms and (**E** to **H**) quantification of GFP expression in αKG-RE–expressing HEK293 cells with EV or one of three VP64-NtcA-2xNLS$_{SV40}$ chimeras: WT NtcA, NtcA^{R90E} mutant, or DNA binding domain (DBD)–deleted NtcA mutant. Cells were cultured under the indicated concentrations of glutamine and/or dm-αKG for 72 hours. In (E) to (H), data are normalized to 2 mM Gln condition. ***$P < 0.001$ (ordinary one-way ANOVA). (**I**) GFP expression normalized to 2 mM Gln condition in HEK293 cells with functional (NtcA^{WT}) or inactive (NtcA^{R90E}) versions of the αKG-ON biosensor system and cultured with indicated concentrations of Gln and/or dm-αKG for 72 hours. **$P < 0.01$ (unpaired *t* test). (**J**) Schema of branched chain amino acid (BCAA) metabolism by BCAT1 and BCAT2 transaminases. BCKA, branched chain α-ketoacid. (**K**) Relative αKG levels in whole-cell extracts by means of liquid chromatography–mass spectrometry (LC-MS) from HEK293 cells with BCAT1 or BCAT2 deletion. **$P < 0.01$ (unpaired *t* test). (**L**) Relative nuclear αKG levels (normalized to sgControl line) in cells from (K) with the functional αKG-ON biosensor system. **$P < 0.01$ (unpaired *t* test). (**M**) Quantification of GFP expression (normalized to sgControl+EV lines) in HEK293 cells with a functional or inactive version of the αKG-ON biosensor system. Cells expressed Cas9 and a control or a BCAT1 sgRNA, as well as an sgRNA-resistant BCAT1 cDNA or EV. (**N**) Immunofluorescence microscopy of HEK293 cells expressing FLAG-tagged BCAT1^{WT} enzyme, BCAT2^{WT} enzyme, or a BCAT1 mutant enzyme with an N-terminal mitochondrial targeting sequence (MTS) peptide. Merge shows overlay of all signals. Scale bar, 10 μm. (**O**) Relative nuclear αKG levels [GFP median fluorescence intensity (MFI) normalized to EV line] in cells expressing the functional αKG-ON biosensor system and engineered as in (N) or expressing an EV. *$P < 0.05$, **$P < 0.01$ (unpaired *t* tests). In (E) to (I), (K) to (M), and (O), data are means ± SEM.

nuclear and cytosolic αKG pools are thought to equilibrate owing to the flow of metabolites and other small molecules through nuclear pores (*35*, *36*). Consistent with this idea, increased activity of the cytosolic branched-chain amino acid transaminase 1 (BCAT1), but not mitochondrial BCAT2, has been shown to cause DNA hypermethylation in leukemia cells (Fig. 2J) (*37*). This interaction was attributed to BCAT1-dependent depletion of nucleocytosolic αKG, which is a substrate for ten-eleven translocation (TET) methylcytosine dioxygenases. We exploited this difference between BCAT paralogs to assess the ability of the αKG-ON biosensor system to monitor compartmentalized αKG metabolism. Deletion of BCAT2 caused a larger increase in whole-cell αKG levels relative to deletion of BCAT1 (Fig. 2K and fig. S5A). Conversely, BCAT1, but not BCAT2, deletion elevated nuclear αKG levels, as indicated by functional biosensor output (Fig. 2L). This effect of BCAT1 deletion was rescued by expression of a single guide RNA (sgRNA)–insensitive BCAT1 cDNA (Fig. 2M and fig. S5, B and C). Moreover, BCAT1 deletion did not affect output of the inactive αKG-ON biosensor system. BCAT1 and BCAT2 paralogs have distinct properties beyond subcellular localization. We next engineered cells to express FLAG-tagged WT BCAT1, WT BCAT2, or a BCAT1 mutant harboring an N-terminal mitochondrial targeting sequence (MTS) (fig. S5D). We confirmed that WT BCAT1 localized to the nucleocytosolic compartment, whereas WT BCAT2 and MTS-BCAT1 enzymes were mitochondrial (Fig. 2N). Overexpression of WT BCAT1 reduced nuclear αKG levels, whereas WT BCAT2 and MTS-BCAT1 did not (Fig. 2O). Thus, the αKG-ON biosensor system senses compartment-specific changes in αKG abundance caused by genetic manipulation of metabolic enzyme function.

A forward genetic approach identified genes that regulate the nuclear αKG pool

We next devised an unbiased forward genetic screen to determine the key molecular regulators of the nuclear αKG pool (Fig. 3A). We used U251 differentiated glioblastoma cells because of their excellent performance in genetic screens (*38*, *39*). We engineered U251 cells with a highly active αKG-RE promoter (tiled NtcA binding sites from the *Synechocystis* sp. PCC 6803 *glnA* promoter with native spacer sequences) (Fig. 1C) and WT or R90E NtcA chimeras. We then introduced Cas9 and a custom "αKG Regulators" CRISPR deletion sgRNA library targeting 127 genes encoding proteins that directly synthesize, catabolize, or transport αKG (data S1). Cells that displayed GFP expression in the top and bottom 10 percentiles of each arm were then isolated by means of fluorescence-activated cell sorting (FACS) and processed for sgRNA quantification. We identified sgRNAs that altered biosensor output in an αKG-dependent manner by surveying patterns of sgRNA enrichment or depletion selectively observed in functional biosensor-expressing cells.

This screen produced four gene-level "hits" (data S2). Deletion of *BCAT1* and *DLD* increased nuclear αKG levels, whereas deletion of *GPT2* and *SLC25A11* depleted this metabolite pool (Fig. 3B). Notably, sgRNAs that targeted these four genes were preferentially enriched or depleted in cells expressing the functional versus inactive version of the biosensor (Fig. 3C, fig. S7A, and data S2). Our findings implicate two mitochondrial proteins, the alanine transaminase GPT2 and the malate/αKG antiporter SLC25A11, as key suppliers of αKG in the nucleus. Conversely, the mitochondrial dihydrolipoamide dehydrogenase DLD and the nucleocytosolic BCAA transaminase BCAT1 represented suppressors of nuclear αKG. Observing *BCAT1* as a top hit suggested that the screen was successful, given previous work (*37*) and our earlier data linking BCAT1 with negative regulation of the nuclear αKG pool (Fig. 2, L and M). We

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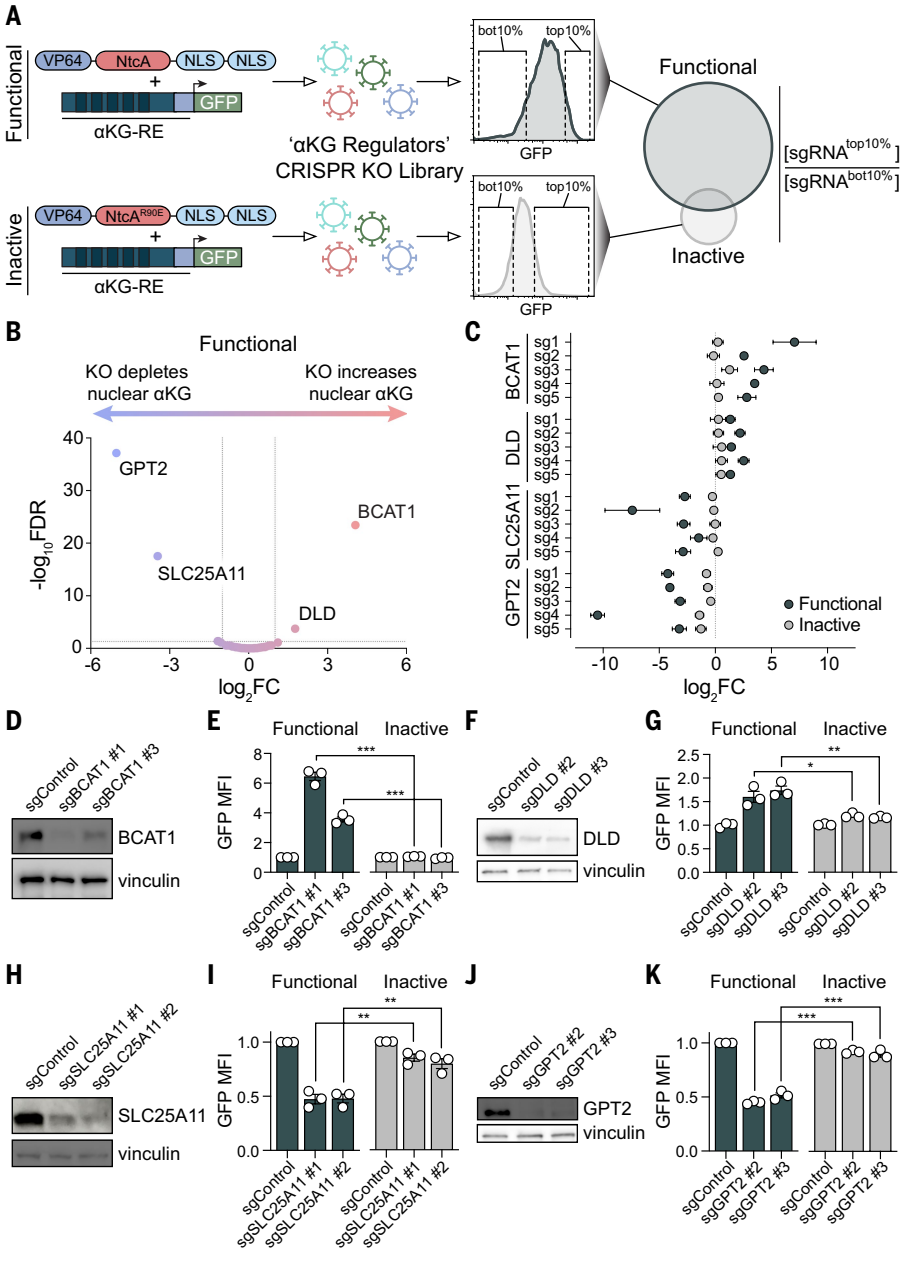


Fig. 3. A forward genetic screen reveals molecular regulators of the nuclear αKG pool. (**A**) Schema of a FACS-based CRISPR-Cas9 deletion screen to identify genes that control nuclear αKG. (**B**) Volcano plot of gene-level statistics from the CRISPR-Cas9 screen in U251 cells engineered with the functional αKG-ON biosensor system shown in (A). FC, fold change; FDR, false discovery rate. Gene-level scores were derived from the ratio of sgRNA read counts in the top 10% versus bottom 10% of GFP-expressing cells. (**C**) Enrichment or depletion of individual sgRNAs targeting BCAT1, DLD, SLC25A11, or GPT2 from CRISPR-Cas9 screens in U251 cells engineered with functional or inactive versions of the αKG-ON biosensor system. (**D** to **K**) Validation of nuclear αKG pool regulation by sgRNAs targeting BCAT1, DLD, SLC25A11, or GPT2. [(D), (F), (H), and (J)] Immunoblots of BCAT1, DLD, SLC25A11, or GPT2 expression in U251 cells engineered to express Cas9 and control or indicated sgRNAs. [(E), (G), (I), and (K)] Quantification of GFP expression (GFP MFI normalized to sgControl lines) in cells from (D), (F), (H), and (J) engineered with functional or inactive versions of the αKG-ON biosensor system. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ (unpaired *t* tests). In (C), (E), (G), (I), and (K), data are means ± SEM.

validated two sgRNAs targeting each of these genes, demonstrating effective depletion of protein products and preferential regulation of functional αKG-ON biosensor system output (Fig. 3, D to K). We also rescued nuclear αKG depletion caused by GPT2 or SLC25A11 deletion by expressing sgRNA-insensitive cDNAs for each target (fig. S6, A to D), validating on-target effects of sgRNAs. Genes that did not score as "hits" in our screen included those encoding the BCAA transaminase BCAT2, glutamate dehydrogenase 1 (GLUD1), the IDH3A subunit of the isocitrate dehydrogenase 3 (IDH3) complex, and the OGDH component of the oxoglutarate dehydrogenase complex (OGDC) (fig. S7B). These enzymes are mitochondrial, suggesting that metabolic reactions in this organelle make variable contributions to the nuclear αKG pool. We validated the inability of BCAT2 to regulate nuclear αKG (fig. S7, C and D), corroborating our previous experiments (Fig. 2L). GLUD1 has a very minor role in controlling nuclear αKG in this model system (fig. S7, E and F). We did observe moderate negative and positive regulation of nuclear αKG upon acute IDH3A and OGDH deletion, respectively (fig. S7, G to J). It may be possible to capture moderate effects such as these in future screens by reducing the stringency of fluorescence intensity thresholds used for cell sorting. Reducing the gating stringency from top and bottom 10 to 30% for *IDH3A* sgRNAs decreased relative variance between replicates (fig. S7, K and L).

We compared nuclear and absolute, whole-cell αKG levels in stable lines with deletion of the aforementioned genes (fig. S8A). Whereas deletion of GPT2, SLC25A11, OGDH, and BCAT1 produced congruent effects on whole-cell and nuclear αKG pools, deletion of DLD, GLUD1, BCAT2, and IDH3A did not. For example, DLD loss increased nuclear αKG but had minimal effect on whole-cell αKG (fig. S8B). These data further highlight ability of the αKG-ON biosensor system to detect compartment-specific changes in αKG abundance that may not be accurately inferred from changes in whole-cell αKG content. We also surveyed how deletion of these genes affected cell proliferation. In general, we observed modest or negligible reductions in cell fitness relative to control sgRNA-expressing cells (fig. S8, C to J). However, one sgRNA targeting OGDH [a common essential gene (*40*)] nearly abolished proliferation (fig. S8K), an effect that correlated with potent OGDH protein depletion (fig. S7I). To ask how sgRNAs targeting essential genes affect biosensor output, we deleted the ribonucleoprotein gene *SNRPG* (small nuclear ribonucleoprotein polypeptide G). SNRPG-deficient lines failed to grow and displayed increased GFP levels in both functional and inactive biosensor-expressing cells (fig. S8, L to N). This effect was partly attributable to autofluorescence of dying cells because SNRPG-deficient cells lacking the GFP reporter construct also displayed higher signal in the GFP channel (fig. S8O). These results helped explain enrichment of toxic OGDH sgRNAs in the GFP^{high} population of inactive biosensor-expressing cells (fig. S7B) and validated our use of the inactive biosensor to deprioritize sgRNAs that cause αKG-independent changes in GFP signal (Fig. 3A). Notably, sgRNAs targeting the top "hits" in our screen (*BCAT1*, *DLD*, *GPT2*, and *SLC25A11*) did not cause gross defects in cell fitness or mitochondrial function (fig. S8, D to G and P), nor did they pervasively increase output of the inactive biosensor (Fig. 3C).

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An interorganelle pathway controls αKG-dependent chromatin demethylation

We hypothesized that the proteins nominated by our screen form a molecular network that dictates the flow of αKG from mitochondria to nuclei (Fig. 4A). We propose that GPT2 contributes a major fraction of the αKG pool that is exported from mitochondria in cells that display robust GPT2-dependent alanine synthesis. Another fraction is controlled by IDH3 and OGDC activities in the tricarboxylic acid (TCA) cycle, as reflected by changes in biosensor output caused by acute IDH3A, OGDH, and DLD deletion (Fig. 3, F and G, and fig. S7, G to J). Although OGDH sgRNAs did not score in the screen (likely owing to chronic toxicity), OGDH deletion did increase nuclear αKG in short-term experiments conducted before OGDH depletion caused loss of fitness (fig. S7, I and J). DLD is the E3 subunit required for generating lipoamide to support catalytic cycles of three mitochondrial enzyme complexes: OGDC, pyruvate dehydrogenase (PDH), and branched chain α-ketoacid dehydrogenase (BCKDH). BCKDH and PDH activities are unlikely to account for DLD-dependent control of the nuclear αKG pool because neither BCAT2 (the upstream regulator of BCKDH) nor PDHA1 (the E1 subunit of PDH) deletion altered functional αKG-ON biosensor system output (figs. S7, C and D, and S9, A and B). DLD likely regulates nuclear αKG through OGDC. DLD loss (Fig. 3, F and G) phenocopies nuclear αKG accumulation caused by acute OGDH deletion (fig. S7, I and J). Considering that both DLD and OGDH are required for OGDC activity, it is not clear why DLD sgRNAs preserved cell viability, whereas OGDH sgRNAs did not (fig. S8, E and K). To complete this interorganelle molecular network, our data suggest that mitochondrial αKG produced by GPT2 or through TCA cycle activity is exported by the SLC25A11 transporter. In the nucleocytosolic compartment, αKG is predominantly catabolized to glutamate by BCAT1 or used as a substrate by Jumonji C domain–containing histone lysine demethylase (KDM) or TET αKG-dependent dioxygenases to promote histone and DNA demethylation, respectively.

To test this model, we first asked whether nuclear αKG depletion caused by GPT2 or SLC25A11 deletion led to impaired histone or DNA demethylation. GPT2 or SLC25A11 deletion caused histone 3 lysine 4 (H3K4) and H3K9 hypermethylation in the presence, but not absence, of ascorbate (Fig. 4B and fig. S9, C to E). A similar trend was observed for H3K27 trimethylation (H3K27me3). In U251 cells, H3K4me3 was most responsive to changes in nuclear αKG, so we preferentially monitored this mark in subsequent in vitro studies. Methylation of deoxycytidine nucleosides in genomic DNA also increased upon GPT2 or SLC25A11 deletion (Fig. 4C), although this effect did not reach statistical significance in genome-wide assays. Supplementing GPT2-null cells with dm-αKG rescued nuclear αKG and H3K4, H3K9, and H3K27 hypermethylation (fig. S9, F to J). Treating GPT2-WT cells with dm-αKG did not cause histone demethylation, suggesting that nuclear αKG is not limiting for activity of KDM enzymes under nutrient-replete conditions. Control of nuclear αKG by GPT2 and SLC25A11 is generalizable because depletion of either protein reduced nuclear αKG in two other neural cell lines: immortalized normal human astrocytes (NHA) astrocytes and

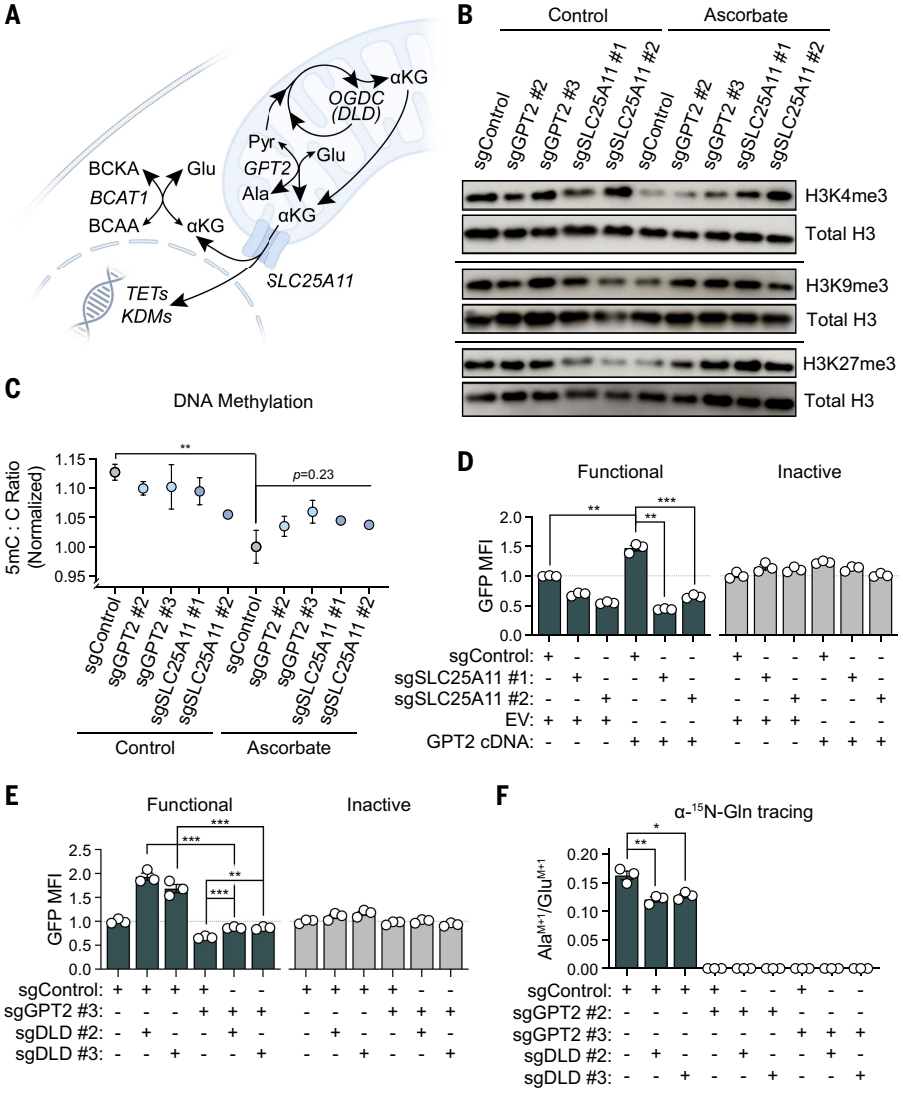


Fig. 4. A mitochondrion-to-nucleus pathway of αKG metabolism controls chromatin methylation. (**A**) Schema of localization patterns and functions of proteins that regulate nuclear αKG. OGDC, oxoglutarate dehydrogenase complex; Pyr, pyruvate; TETs, ten-eleven translocation methylcytosine dioxygenases; KDMs, Jumonji C domain-containing histone lysine demethylases. (**B**) Immunoblots of histone H3 total protein levels or trimethyllysine posttranslational modifications in U251 cells engineered to express Cas9 and control, GPT2, or SLC25A11 sgRNA. Cells were treated with or without 200 μM ascorbate-2-phosphate. (**C**) Ratios of 5-methyldeoxycytidine (5mC) to deoxycytidine (C) in hydrolyzed genomic DNA isolated from cells in (B). Ratios are normalized to ascorbate-treated sgControl line. $n = 4$ biological replicates per condition. $^{**}P < 0.01$ (unpaired t test), $P = 0.23$ (ordinary one-way ANOVA). (**D**) Quantification of GFP expression (GFP MFI normalized to sgControl+EV lines) in U251 cells engineered to express Cas9 and control or SLC25A11 sgRNA, as well as GPT2 cDNA or EV. Cells were also engineered with functional or inactive versions of the αKG-ON biosensor system. $^{**}P < 0.01$, $^{***}P < 0.001$ (unpaired t tests). (**E**) Quantification of GFP expression (GFP MFI normalized to sgControl lines) in U251 cells engineered to express Cas9 and control, GPT2, or DLD sgRNA. Cells were also engineered with functional or inactive versions of the αKG-ON biosensor system. $^{**}P < 0.01$, $^{***}P < 0.001$ (unpaired t tests). (**F**) α-^{15}N-Glutamine stable isotope tracing for 2 hours in U251 cells engineered to express Cas9 and control, GPT2, or DLD sgRNA. Fractional enrichment of the Ala (M+1) isotopologue relative to fractional enrichment of the Glu (M+1) isotopologue is shown for each line. $^{*}P < 0.05$, $^{**}P < 0.01$ (unpaired t tests). In (C) to (F), data are means ± SEM.

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TS516 patient-derived glioma stem-like cells (fig. S10, A to D). GPT2 deletion in TS516 cells also led to nuclear αKG depletion when these cells were cultured with physiological concentrations of alanine and glutamine and subjected to variations in glucose, pyruvate, and oxygen (fig. S10E). Therefore, GPT2 supplies αKG to the nucleus across metabolic stress conditions that are relevant in vivo.

Mitochondrial GPT2 and SLC25A11 activities supply the nuclear αKG pool

The functions of GPT2 and SLC25A11 implied that they act in a linear pathway to produce and export αKG, respectively, from mitochondria. GPT2 overexpression increased αKG levels in the nucleus in an SLC25A11-dependent manner (Fig. 4D and fig. S12A). Additionally, GPT2 upregulation did not rescue histone hypermethylation caused by SLC25A11 deletion (fig. S12, B to E). To assess subcellular distribution of GPT2 activity, we first quantified GPT2 splice isoforms endogenously expressed by U251 and NHA cells (fig. S11, A and B). In both cell types, the predominant transcript expressed, ENST00000340124.9, encodes full-length GPT2 protein with a predicted N-terminal MTS (fig. S11, C and D). However, the second most abundant transcript expressed in both cell lines, ENST00000440783.2, features a distinct 5′ exon structure and encodes a GPT2 protein isoform that lacks an MTS. Overexpression of the two dominant GPT2 transcripts in U251 cells revealed the protein product of ENST00000440783.2 clearly localized to the nucleocytosolic compartment, in contrast to the full-length ENST00000340124.9 (fig. S11, E and F). Overexpression of the truncated GPT2 enzyme had a negligible effect on nuclear αKG (fig. S11G). Thus, αKG produced in mitochondria by GPT2 is exported by SLC25A11 to supply the pool of αKG in nuclei.

We next sought to investigate the interaction between GPT2 and DLD. Both GPT2-WT and GPT2-null cells displayed increased nuclear αKG upon DLD deletion (Fig. 4E and fig. S12, F and G), and DLD deletion rescued histone hypermethylation caused by GPT2 loss (fig. S12, H to K). Thus, DLD-dependent regulation of αKG signaling to chromatin does not require GPT2. Deletion of OGDH also increased output of the functional αKG-ON biosensor system independent of GPT2 (fig. S12, L and M), albeit with confounding effects on inactive biosensor output due to loss of cell fitness. We next performed stable isotope tracing with α-^{15}N-glutamine in control cells or those lacking GPT2, DLD, or both enzymes to determine the effect of DLD loss on GPT2 flux, using the ratio of alanine labeling to glutamate labeling as a marker of GPT2 activity (Fig. 4F and fig. S12N). DLD deletion reduced GPT2 flux in the direction of pyruvate catabolism, and GPT2 deletion abolished alanine labeling by α-^{15}N-glutamine. Thus, GPT2 and DLD occupy critical but distinct, nonoverlapping nodes within the nuclear αKG regulatory network.

A mouse model of GPT2 deficiency displays chromatin hypermethylation in the brain

We evaluated a mouse model of GPT2 deficiency to determine the epigenetic regulatory role of GPT2 in vivo. GPT2 deficiency is a relatively recently described inborn error of metabolism caused by autosomal recessive inheritance of loss-of-function mutations in the *GPT2* gene (*41*–*44*). These mutations cause neurological and intellectual impairment. In mice, whole-body Gpt2 deletion leads to death 17 to 30 days after birth (*45*). This phenotype is fully recapitulated by neuron-specific deletion of Gpt2, underscoring a critical function for this enzyme in supporting neuronal development and homeostasis (*46*). We surveyed global histone methylation profiles of various tissues from whole-body Gpt2-deletion [Gpt2 KO (knockout)] and Gpt2 WT mice and observed pervasive H3K4, H3K9, and H3K27 hypermethylation in the brains of Gpt2 KO animals at postnatal day 19 (P19) (Fig. 5A and fig. S13, A and H to J). There was generally little effect of GPT2 loss on histone methylation in other organs, aside from hypermethylated H3K9 in heart tissues and hypermethylated H3K9 and H3K27 in spleen tissues (fig. S13, B to G). We also observed a trend toward global 5mC enrichment in genomic DNA extracts from Gpt2 KO versus Gpt2 WT mouse brain tissues (Fig. 5B). This trend was specific to brain because Gpt2 loss did not alter genomic 5mC levels in kidney tissues (Fig. 5C). Performing genome-wide analysis of 5mC with single-base resolution, we observed a significant increase in DNA methylation in Gpt2 KO versus Gpt2 WT brain tissues (Fig. 5D). Thus, Gpt2 loss causes chromatin hypermethylation in the brain.

Pyruvate is enriched in cerebrospinal fluid and dictates GPT2-dependent αKG synthesis

We hypothesized that tissue-specific differences in the levels of GPT2 substrates, rather than differences in GPT2 expression, determine the directionality and magnitude of GPT2 flux. GPT2 is reversible and can also catabolize αKG and alanine, yielding pyruvate and glutamate. Alanine levels are reduced in brain tissues of Gpt2 KO mice (*42*), suggesting that net flux through GPT2 likely moves in the direction of αKG and alanine synthesis in brain tissue in vivo. Thus, we asked whether increased pyruvate availability may be a driver of GPT2-dependent αKG and alanine synthesis in the central nervous system (CNS). The pyruvate content of cerebrospinal fluid (CSF) is nearly twofold greater than that of blood in both children and adults (fig. S13, K and L), suggesting that the local supply of pyruvate may be greater in the CNS than in peripheral organs. To determine whether pyruvate availability affects GPT2-dependent supply of the nuclear αKG pool, we evaluated α-^{15}N-glutamine stable isotope tracing and αKG-ON biosensor system output in control or GPT2 KO cells cultured in medium with or without pyruvate. Pyruvate supplementation stimulated GPT2 flux in the direction of alanine and αKG synthesis (fig. S13, M and N) and markedly elevated nuclear αKG (fig. S13O). GPT2 deletion abolished both effects. Pyruvate supplementation caused GPT2-dependent trends toward H3K4, H3K9, and H3K27 hypomethylation (fig. S13, P to S). Thus, the preferential epigenetic regulatory function of GPT2 in the CNS may be driven by high microenvironmental pyruvate concentrations.

Gpt2 loss dysregulates chromatin methylation and neurodevelopmental gene programs in an αKG-dependent manner

Most research on the molecular pathogenesis of GPT2 deficiency has focused on defects in alanine biosynthesis, glutamine-dependent TCA cycle anaplerosis, and glutamatergic neurotransmission (*42*, *47*). Our work suggests that depletion of nuclear αKG and chromatin hypermethylation may contribute to disease phenotypes as well. To test the hypothesis that nuclear αKG depletion causes the chromatin changes seen in Gpt2 KO mouse brains, we administered dm-αKG to Gpt2 WT and Gpt2 KO mice. Treatment with dm-αKG rescued histone hypermethylation caused by Gpt2 loss (Fig. 5, E to H) and preferentially affected gene expression in brain tissues of Gpt2 KO mice relative to that of Gpt2 WT controls (Fig. 5, I to K, and data S6). dm-αKG supplementation also rescued weight loss in Gpt2 KO mice that precedes their early postnatal lethality (Fig. 5L and fig. S13T) (*45*). Thus, defects in αKG signaling to chromatin appear to play a central, causal role in the molecular pathogenesis of GPT2 deficiency.

We also evaluated the epigenetic regulatory role of GPT2 in tumors. *PIK3CA* mutations increase GPT2 activity in cancer cells (*48*). Using *PIK3CA*-mutant HCT116 colorectal cancer cells, we generated isogenic GPT2 WT and GPT2 KO stable lines and implanted them subcutaneously into immunodeficient mice. We treated mice with dm-αKG or vehicle and subsequently harvested tumors for histone methylation analysis. GPT2 loss caused H3K4, H3K9, and H3K27 hypermethylation, and these effects were rescued by means of dm-αKG treatment (fig. S14, A to D). Vehicle-treated GPT2 WT and GPT2 KO tumors grew at similar rates, indicating that histone hypermethylation is not caused by nonspecific loss of cancer cell fitness in GPT2-deficient tumors (fig. S14E). These data indicate that GPT2-dependent control of chromatin methylation occurs in diverse biological contexts.

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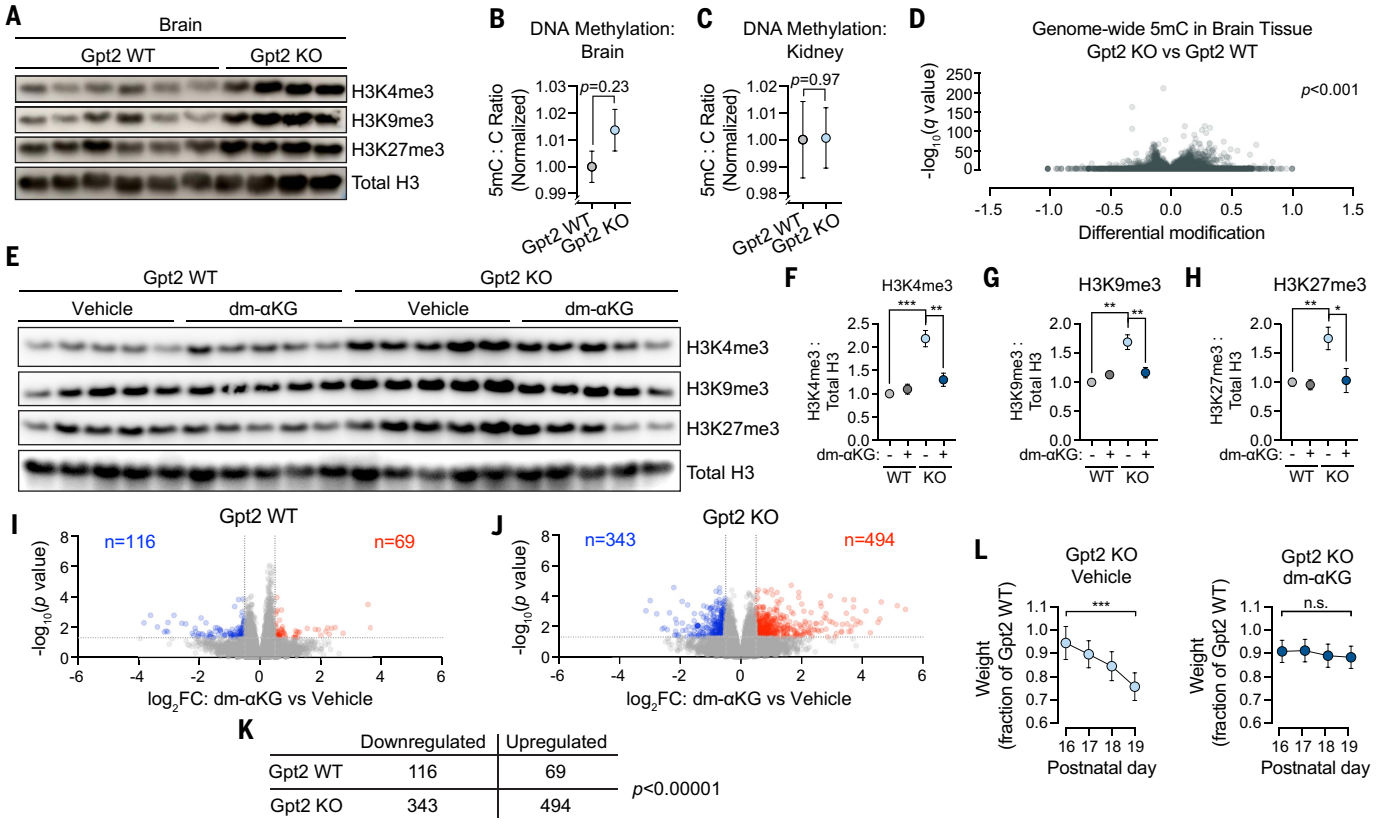


Fig. 5. Gpt2 loss causes αKG-dependent epigenetic dysregulation and reduced fitness in mice. (**A**) Immunoblots from brain tissues of Gpt2 WT or Gpt2 KO mice. (**B** and **C**) 5mC-to-deoxycytidine ratio in hydrolyzed genomic DNA isolated from (B) brain or (C) kidney tissues from Gpt2 WT or Gpt2 KO mice. $n = 3$ tissues per arm. *P* values are from unpaired *t* tests. (**D**) Genome-wide, locus-specific analysis of 5mC levels in brain tissues from Gpt2 WT and Gpt2 KO mice. *P* value is from Wilcoxon signed rank test. (**E**) Immunoblots from brain tissues of Gpt2 WT or Gpt2 KO mice treated with vehicle or 600 mg kg^{-1} dm-αKG. (**F**) H3K4me3, (**G**) H3K9me3, and (**H**) H3K27me3 quantification from blots in (E). Normalized intensities were expressed relative to the mean of vehicle-treated Gpt2 WT samples. $n = 5$ brain tissues per arm. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ (unpaired *t* tests). (**I** and **J**) Volcano plots from RNA-seq analysis of brains from (I) Gpt2 WT mice and (J) Gpt2 KO mice treated with vehicle or 600 mg kg^{-1} dm-αKG. (**K**) Differentially expressed genes from (I) and (J). *P* values were calculated by means of χ^2 test. (**L**) Weights of Gpt2 KO mice treated with vehicle ($n = 10$ mice) or 600 mg kg^{-1} dm-αKG ($n = 9$ mice), normalized to Gpt2 WT mice. $^{***}P < 0.001$ (paired *t* test). In (B), (C), (F) to (H), and (L), data are means ± SEM.

We next performed chromatin immunoprecipitation followed by sequencing (ChIP-seq) for H3K4me3, H3K9me3, and H3K27me3 marks in chromatin from Gpt2 KO and Gpt2 WT mouse brains. In agreement with immunoblot analyses of global H3K4me3, H3K9me3, H3K27me3 (Fig. 5A), these marks were enriched at loci throughout the genome in brain tissues from Gpt2 KO mice (Fig. 6, A to C, and data S3 to S5). H3K27me3 marks are well established to suppress transcriptional output. To identify genes that display both H3K27 hypermethylation and decreased expression in the brains of Gpt2 KO mice, we integrated RNA-sequencing (RNA-seq) data from the brains of Gpt2 KO and WT mice (data S6) with H3K27me3 ChIP-seq data (Fig. 6D and data S5). More than 800 genes down-regulated in Gpt2 KO mouse brain tissues also displayed H3K27 hypermethylation, identifying a subset of transcriptional changes that are linked to nuclear αKG depletion.

Gene ontology (GO) analysis highlighted terms related to nervous system development, neurogenesis, and gliogenesis (Fig. 6E and data S7), suggesting that H3K27me3 hypermethylation affects gene programs related to postnatal nervous system development. In addition, genes related to intracellular calcium regulation, a key effector of neuronal electrochemical signaling, and locomotory behavior were substantially altered. Furthermore, the H3K27me3-associated high-CpG-density promoters (HCP) and cranial nerve development gene sets were decreased in Gpt2 KO mouse brains relative to Gpt2 WT and rescued with dm-αKG treatment (Fig. 6, F to I, and data S6). We surveyed three genes with key roles in postnatal CNS development that are dysregulated by Gpt2 loss and rescued with dm-αKG treatment (Fig. 6, J to O). *Hapln4* encodes an extracellular matrix protein that stabilizes synaptic perineuronal nets (*49*). Hapln4 down-regulation is associated with impaired inhibitory neurotransmission (*50*) and neurodegenerative processes in aging and stroke (*51*). *Etv4* encodes a transcription factor (*52*, *53*) required for development of motor and sensory neuron circuits (*54*, *55*). *Notch3* is expressed in cerebral vascular smooth muscle cells and neural progenitor cells and causes decreased blood flow and aberrant neuronal differentiation when inactivated (*56*–*58*).

The H3K4me3 mark is thought to play a stabilizing but not driving role in gene transactivation (*59*). More than 1200 genes were H3K4me3 hypermethylated and overexpressed in Gpt2 KO mouse brains (fig. S15A and data S3 and S6), including many related to amino acid and nutrient starvation, synaptic translation, cognition, and neuropeptide signaling (fig. S15B and data S8). We surveyed three of these genes, including *Eif4ebp1*, which is stimulated by the integrated stress response (*60*); *Cux2*, which encodes a transcription factor that controls neurogenesis and synaptogenesis (*61*, *62*); and *Nrg3*, which encodes a neuregulin family member that regulates cortical interneuron migration (*63*). Aberrant up-regulation of each of these genes was suppressed with dm-αKG treatment (fig. S15, C to H). These data establish a key epigenetic regulatory function for GPT2 in vivo and reveal epigenetic dysfunction as a contributor to defective neurodevelopment in GPT2 deficiency.

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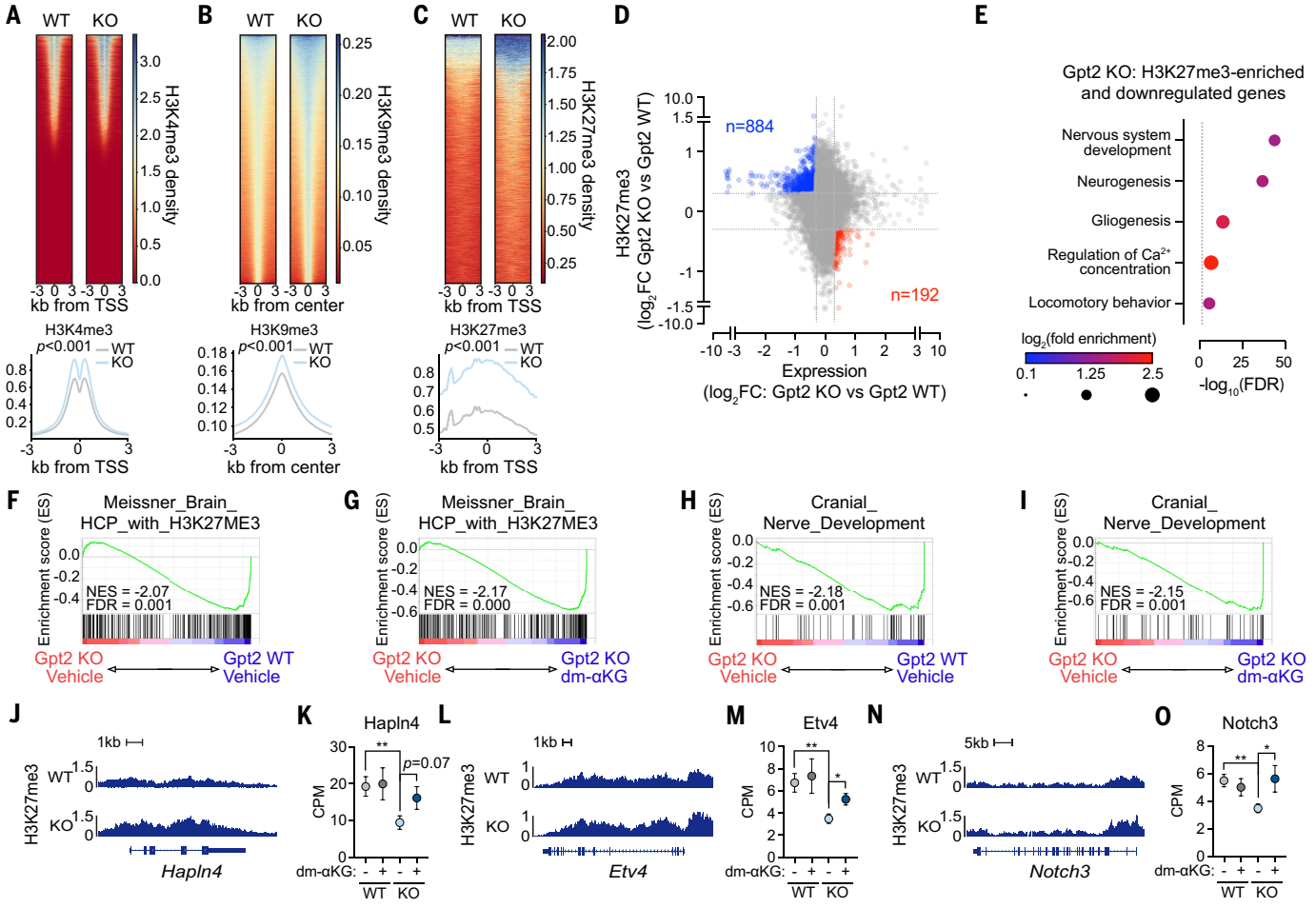


Fig. 6. αKG-dependent neurodevelopmental gene dysregulation occurs in Gpt2 KO mice. (**A**) H3K4me3, (**B**) H3K9me3, or (**C**) H3K27me3 ChIP-seq results from Gpt2 WT or Gpt2 KO mice brains. Tornado plots and quantification of aligned histone methylation densities. (**D**) Integrative analysis of H3K27me3 ChIP-seq and RNA-seq datasets from brain tissues of Gpt2 WT and Gpt2 KO mice. (**E**) Gene ontology analysis of genes derived from analysis in (D). (**F** to **I**) Enrichment plots for [(F) and (G)] genes with high-CpG-density promoters (HCP) bearing the H3K27me3 mark in brain and [(H) and (I)] cranial nerve development genes. NES, normalized enrichment score. (**J**, **L**, and **N**) Density tracks for H3K27me3 ChIP-seq for H3K27me3 hypermethylated and down-regulated neurodevelopmental genes in Gpt2 KO mouse brains. (**K**, **M**, and **O**) RNA-seq counts per million (CPM) expression plots are included for indicated genes from Gpt2 KO and Gpt2 WT mice treated with or without dm-αKG. $n = 10$ brain tissues for vehicle-treated Gpt2 WT and Gpt2 KO mice and $n = 8$ brain tissues for Gpt2 WT and Gpt2 KO mice treated with dm-αKG. $^*P < 0.05$, $^{**}P < 0.01$ (unpaired t tests). In (K), (M), and (O), data are means ± SEM.

Discussion

In this work, we introduced the αKG-ON biosensor system as a tool that enables robust, specific measurement of changes in nuclear αKG pool size. Our work expands on techniques available to measure αKG in live cells. Previous fluorescence resonance energy transfer (FRET)–based tools were largely used to monitor αKG in bacteria (*33*, *64*–*69*). The αKG-ON biosensor system may be advantageous for monitoring αKG in human cells when compartmental specificity, high sensitivity, and enhanced stability of αKG sensing are desired. We determined the K_d of NtcA binding to αKG to be 2.0 mM and whole-cell αKG content of U251 cells to be 1.35 mM. Accordingly, the biosensor detected changes in nuclear αKG abundance when whole-cell αKG varied from ~0.5 to 6 mM. Michaelis constant (K_m) values for αKG binding to TET and KDM enzymes are 5 to 60 μM (*70*), implying that nuclear αKG may not be limiting for their activities. However, these K_m values were derived from studies of recombinant, purified enzymes in the absence of competing inhibitory molecules, such as succinate and fumarate (*4*). Our work establishes that nuclear αKG is not limiting for chromatin demethylation under basal conditions but becomes limiting when the enzymes and transporters that supply this metabolite pool are impaired.

Results from our biosensor-enabled forward genetic screen informed a conceptual model of nuclear αKG regulation. GPT2 and TCA cycle activities regulate αKG produced in mitochondria that is subsequently exported to the cytosol through SLC25A11. αKG may then diffuse into nuclei through nuclear pores (*35*, *36*). The nuclear αKG pool sustains dioxygenases involved in chromatin demethylation and is antagonized by BCAT1, which is in agreement with prior research (*37*). Supporting the physiological relevance of this model, GPT2 activity is critical for establishing appropriate chromatin methylation patterns in the brain.

Our findings suggest that nuclear αKG in human cells is controlled differently than the nuclear pools of acetyl-CoA and SAM, which also serve as substrates for chromatin-modifying enzymes. Adenosine 5′-triphosphate (ATP)–citrate lyase (ACLY) (*71*, *72*), acetyl-CoA synthetase 2 (ACSS2) (*73*–*75*), pyruvate dehydrogenase (PDH) (*76*, *77*), and methionine adenosyltransferase (MAT) (*78*–*80*) have been shown to localize to the nucleus to facilitate chromatin modification. By contrast, we found that nuclear αKG is principally derived from mitochondrial metabolism. GPT2 has been reported to localize to the nucleus (*81*), but we observed that mitochondrial GPT2 activity supplies αKG to nuclei. We also showed that SLC25A11 is epistatic to GPT2 in an

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interorganelle pathway that sustains the nuclear αKG pool. This finding aligns with a study of familial paraganglioma that demonstrated that *SLC25A11* loss-of-function mutations phenocopy chromatin hypermethylation caused by oncometabolite-producing *SDH* or *FH* mutations (*82*).

Our work provides insights into the pathogenesis of GPT2 deficiency. Symptoms of this disorder include global developmental delay, hypotonia, delays or failure to develop speech and walking, and epilepsy (*41*–*44*). Loss of GPT2-dependent alanine synthesis is implicated in these disease-related phenotypes because alanine supplementation improves the health of cultured neurons and mice lacking Gpt2 (*45*). Our findings revealed loss of GPT2-dependent αKG synthesis to be another contributor to neurological dysfunction in GPT2 deficiency. Genes related to neurogenesis and synaptogenesis were dysregulated in Gpt2 KO mice. αKG supplementation normalized these effects and prevented weight loss that precedes lethality in these animals. Our study establishes that nuclear αKG pool depletion and epigenetic reprogramming drive the pathology of GPT2 deficiency and that αKG supplementation may represent a promising strategy to treat this disorder.

Materials and methods

Experimental model details

Mice: All experiments involving live mice were carried out in accordance with the National Institutes of Health *Guide for the Care and Use of Laboratory Animals* under the protocols approved by the Brown University Institutional Animal Care and Use Committee (24-09-0009) and UT Southwestern Medical Center (2019-102795) Institutional Animal Care and Use Committee. Animal welfare assessments were carried out daily during treatment periods. All care and treatment of experimental animals were carried out in strict accordance with Good Animal Practice as defined by the US Office of Laboratory Animal Welfare.

Gpt2 cryopreserved mutant embryos were established through resources at the Knockout Mouse Project at University of California, Davis (Project ID CSD24977). The background of the embryos is C57BL/6N, and these mice were fully backcrossed into a C57BL/6J line. The targeting construct involved a gene trap (splice acceptor) and *LacZ*-Neo cassette (*42*). The mice under the C57BL/6J background were then fully backcrossed onto the CD-1 line. Heterozygous males and heterozygous females were mated to produce offspring of both genders and all genotypes. Animals were genotyped by PCR with forward primer GPT2 neF (5′– TGATGGCACTCTGCACCTAC-3′) and reverse primer GPT2 neR (5′–TCACTCTTGGCTCTGGACCT-3′) for a WT band of 830 bp and a mutant band of 900 bp, as described previously (*42*). Mice were maintained on standard chow. Where indicated, mice were treated once daily with vehicle or 600 mg kg^{-1} dm-αKG (Sigma 349631) by intraperitoneal injection starting on postnatal day 0 (P0). On postnatal day 19 (P19), tissues including brain, heart, liver, kidney, lungs, spleen, stomach, colon, and sciatic nerve were harvested. All tissues were immediately snap-frozen and subsequently stored in liquid nitrogen until further analysis.

NCR/NU mice (Taconic) were used for subcutaneous xenograft experiments. Tumor cells were injected into female mice at 6 weeks of age, and experimental groups were matched for age.

Cell lines: NHA cells (human astrocytes immortalized with HPV E6 and E7 and hTERT, sex unknown) were a kind gift of R. Pieper (UCSF) (*83*). HEK293 cells (female) were purchased from ATCC (CRL-1573). HEK293T cells (female, transduced with SV40 T-antigen) were a kind gift of W. G. Kaelin Jr. (Dana-Farber). U251 cells (male) were purchased from Sigma (09063001-1VL). HOG cells (human oligodendroglioma line from a male) were a gift of P. Paez (SUNY Univ. at Buffalo). Human TS516 glioblastoma stem-like cells (GSCs) (sex unknown) were obtained from I. Mellinghoff (MSKCC) and reported previously (*84*). HCT116 cells (male) were a kind gift of J. Garcia-Bermudez (UTSW).

Unless otherwise indicated, all cells were cultured at 37°C, in the presence of ambient oxygen and 5% CO_2. HEK293, HEK293T, U251, NHA, and HCT116 cells were cultured in DMEM (Gibco 11995065) with 10% FBS and 1% penicillin/streptomycin unless otherwise stated. HOG cells (human oligodendroglioma line from a male) were cultured in IMDM medium (Gibco 12440061) with 10% FBS and 1% penicillin/streptomycin. TS516 cells were cultured in NeuroCult NS-A Basal Medium (Human) with Proliferation Supplement (StemCell Technologies 05750) supplemented with EGF (20 ng/ml), bFGF (10 ng/ml), Heparin (2 μg/ml), 1% penicillin/streptomycin, amphotericin B (250 ng/ml), and Plasmocin (2.5 μg/ml). Where indicated, HOG cells were treated with 250 nM vorasidenib (Wuxi Apptec) for 48 hours. Where indicated, TS516 cells were placed in a modular hypoxia incubator chamber (Baker Ruskinn InvivO$_2$) that was flushed with 1% O_2, 5% CO_2, and 94% N_2.

Methods

NtcA binding site prediction: Previously defined gene-specific NtcA binding sites were compared with the consensus NtcA binding site. The PRODORIC database of prokaryotic gene regulation was used to construct a position weight matrix defining the prevalence of nucleotides at each position within conserved NtcA binding sites derived from various cyanobacterial genomes. The PRODORIC database was accessed at https://www.prodoric.de.

Nuclear export signal prediction: The NetNES1.1 tool (*85*) was used to evaluate putative nuclear export signals (NES) throughout the primary amino acid sequence of the VP64-NtcA-NLS$_{c\text{-}myc}$ NtcA chimera. The NetNES1.1 tool was accessed at https://services.healthtech.dtu.dk/services/NetNES-1.1.

Mitochondrial targeting sequence prediction: Prediction of mitochondrial targeting sequences in protein products of GPT2 splice isoforms was conducted using TargetP-2.0. The TargetP-2.0 tool was accessed at https://services.healthtech.dtu.dk/services/TargetP-2.0.

NtcA structural analysis: Structural data describing the interaction between NtcA and DNA were derived from a cryo-electron microscopy (cryo-EM) study of the cyanobacterial transcription activation complex involving NtcA and NtcB (Protein Data Bank accession 8H3V). The NtcA-DNA complex was isolated from the larger transcription activation complex cryo-EM structure using Pymol software (Schrödinger).

Protein purification and microscale thermophoresis: NtcA was prepared by GenScript in Storage Buffer (100 mM HEPES, pH 7.4, 500 mM NaCl, 0.2 mM EDTA, 0.005% Nonidet P-40, 10% Glycerol). MST Buffer was prepared similarly to Storage Buffer but containing only 320 mM NaCl and no Glycerol. NtcA was concentrated to 20 μM (measured by using extinction coefficient 803,210 $M^{-1}cm^{-1}$ at 205 nm) and fluorescently labeled by mixing 200 μl of the 20 μM solution to 1 μl of diluted Cyanine-5 NHS dye (a mixture of 1 μl of 40 mM dye in DMSO and 2 μl of MST Buffer). The labeling reaction was incubated at room temperature in the dark for 20 min and centrifuged in a microcentrifuge at maximum speed for 10 min at 4°C to remove insoluble material. Free dye was removed using a GE PD Miditrap G-25 column equilibrated with 9 ml of MST Buffer. The eluted Cyanine-5-labeled NtcA was centrifuged again for 10 min at maximum speed in a microcentrifuge to remove residual turbidity. Both αKG (Sigma-Aldrich K1128) and oxaloacetate (OAA) (Sigma-Aldrich O7753) were weighed (with precision to 0.1 mg) and dissolved in MST Buffer to prepare stock solutions of 100 mM. MST experiments on NtcA with titrating αKG were conducted in triplicate, and a single titration was performed with OAA. To set up an experiment, a 20 μl aliquot of 100 mM αKG or OAA in MST buffer was serially diluted 1:1 into 14 additional tubes (i.e., 10 μl was taken from the first tube and mixed with the 10 μl MST Buffer in the second tube, and this process was repeated). The sixteenth tube contained only 10 μl of MST Buffer (no ligand). To each of these sixteen samples, 10 μl of 100 nM labeled NtcA in MST Buffer was added, yielding a final concentration

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of 50 nM for NtcA, and a top concentration of αKG or OAA at 50 mM. The mixtures were incubated in the dark at room temperature for at least 30 min before the commencement of the thermophoretic experiment in the premium capillaries. Experiments were conducted at ambient temperature (approximately 22°C) in a NanoTemper NT.115 BLUE/RED instrument using the red filter set, 7% LED power, and 80% MST power. The pre-IR fluorescence period was 5 s, followed by a 30 s IR-on phase, and 5 s of post-MST monitoring. All MST data were analyzed in PALMIST (*86*), and figures were rendered in GUSSI (*87*).

Cloning: All synthetic DNA sequences, DNA plasmids, and CRISPR-Cas9 sgRNA sequences are listed in tables S2 to S6. In-Fusion cloning reactions were transformed into XL10-Gold Ultracompetent Cells (Agilent Technologies) following manufacturer's protocol. Gateway cloning reactions were performed with BP Clonase II (Invitrogen) or LR Clonase II (Invitrogen) enzymes. Gateway cloning reactions were transformed into HB101 Competent Cells (Promega). Ligation reactions were transformed into XL10-Gold cells. PCR reactions were performed using KOD Xtreme Hot Start DNA Polymerase (EMD Millipore).

αKG-RE promoter/GFP reporter gene lentiviral vectors: First, sense and antisense ssDNAs comprising αKG-RE synthetic promoters were annealed to produce dsDNAs. In parallel, a blasticidin^{resistance}-P2A-GFP cDNA flanked 5′ by a minimal CMV promoter was cloned into pEN_TTmcs (Addgene 25755) via In-Fusion cloning (In-Fusion HD Cloning Kit, Takara 639650) to produce the vector pEN_TTmcs-P_{CMVmin}- blast^{R}-P2A-GFP. Next, In-Fusion cloning was used to insert αKG-RE sequences into pEN_TTmcs-P_{CMVmin}- blast^{R}-P2A-GFP while removing the Tetracycline Response Element (TRE) from this vector. The resulting plasmids were referred to as pEN_TTmcs-[αKG-RE]-P_{CMVmin}- blast^{R}-P2A-GFP.

To accommodate αKG-RE promoter/GFP reporter gene expression in settings in which preservation of cellular blasticidin sensitivity was desired, the blast^{R}-P2A cassette 5′ to the GFP cDNA was deleted by In-Fusion cloning, yielding pEN_TTmcs-[αKG-RE]-P_{CMVmin}-GFP vectors. For experiments testing the impact of GFP turnover on the dynamic range of the αKG-ON biosensor system, an hPEST domain was appended 3′ to the GFP reporter, yielding pEN_TTmcs-[αKG-RE]-P_{CMVmin}- blast^{R}-P2A-GFP-hPEST vectors.

To create a promoterless Gateway lentiviral Destination vector, the PGK promoter was removed from pLenti PGK Neo DEST (w531-1) (Addgene 19067) by In-Fusion cloning to produce the vector pLenti Neo DEST. Finally, Gateway LR cloning reactions (Invitrogen Gateway LR clonase, Fisher Scientific 11-791-100) were performed between αKG-RE/GFP reporter gene-containing pEN_TTmcs vectors and the pLenti Neo DEST vector to generate lentiviral vectors for human cell transduction.

Lentiviral vectors for cDNA expression: Gateway destination vectors used for expression of NtcA chimeras and other cDNAs include: pLenti-Ubc-HA-Gate-PGK-HYG (Addgene 107396), pLenti PGK Hygro DEST (w530-1) (Addgene 19066), pLenti-EF1α-Gateway-PGK-Hygro (kind gift of W. G. Kaelin Jr.), and pLenti CMV Hygro DEST (w117-1) (Addgene 17454).

CRISPR-Cas9 lentiviral vectors: Cloning of sgRNAs (listed in table S5) into vectors was performed using FastDigest BsmBI (Esp3I) (Thermo Scientific) to digest plasmids. Oligonucleotides were annealed and phosphorylated using T4 PNK (NEB) in T4 Ligation Buffer (NEB). A ligation reaction with digested lentiGuide-puro (Addgene 52963), lentiCRISPR_v2 (Addgene 52961), lentiCRISPR_v2-blast (Addgene 83480), or lentiCRISPRv2-mCherry (Addgene 99154) plasmids and the phosphorylated oligonucleotides was performed using Quick Ligase (NEB).

Stable cell line generation: Lentivirus was made using TransIT-LT1 Transfection Reagent (Mirus Bio MIR 2304). HEK293T cells were transfected by mixing TransIT-LT1 Transfection Reagent with expression vectors (listed in table S4) and packaging plasmids psPAX2 (Addgene 12260) and pMD2.G (Addgene 12259) in a ratio of 4:3:1. Virus-containing media were collected 48 and 72 hours after transfection and passed through a 0.45 μm filter to remove cell debris. Cells to be transduced were plated at a density of 150,000 to 300,000 cells per well in a 6-well plate. The next day, lentivirus was added with polybrene (MedChem Express HY-112735) at a final concentration of 8 μg/ml. Cells were then incubated with lentivirus overnight before discarding lentivirus-containing medium and expanding the cells.

Antibiotic concentrations for selection of lentivirus-transduced cell cultures were as follows: puromycin (1 to 2 μg/ml), hygromycin (500 μg/ml), G418 (1 mg/ml), blasticidin (50 μg/ml), zeocin (50 μg/ml).

αKG-ON biosensor system expression: DNA plasmids used to express NtcA chimera and αKG-RE/GFP reporter components of the αKG-ON biosensor system in each figure panel are listed in table S6. Lipofectamine 2000 Transfection Reagent (Invitrogen 11668027) was used for transient transfection of NtcA chimeras. Stable cell lines were generated as described above.

CRISPR screening and analysis: U251 cells expressing the αKG-ON biosensor system were transduced with a Cas9 plasmid with a zeocin resistance marker. Lentivirus containing the custom "αKG Regulators" CRISPR-Cas9 library (data S1) was infected at a multiplicity of infection (MOI) of ~0.3. 3E6 cells were infected per arm of the screen, with cells plated at a density of 1.5E6 cells per well of a 6-well plate. After transduction, cells were split into three replicates per arm and selected with puromycin. One week post-transduction, cells were plated at 250,000 cells per well (1.5E6 total per replicate per arm), and media was changed daily until day ten post-transduction, when cells were harvested for FACS. For each replicate of each arm, the top 10% highest GFP expressing cells, as well as the bottom 10% of GFP expressing cells, were sorted into separate tubes.

Genomic DNA was extracted from cells using the Blood & Tissue DNeasy kit (Qiagen). sgRNA cassettes were amplified and appended to Illumina primers by PCR using KOD Xtreme Hot Start DNA Polymerase (EMD Millipore). Amplicons were purified using SPRI Right Side Size Selection using a 0.7× ratio. Library QC was performed using a TapeStation instrument (Agilent). Library DNA content was quantified using a Qubit fluorometer (Thermo Fisher Scientific) before library sequencing. The sequencing results were then analyzed by MAGeCK (*88*) and Apron (Genetic Perturbation Platform, Broad Institute of MIT and Harvard).

Immunoblotting: Cells were lysed with EBC lysis buffer with protease inhibitor added (Roche). Nuclear and cytoplasmic fractionation was performed using the NE-PER Nuclear and Cytoplasmic Extraction Reagents (Thermo Scientific 78833). For histone Western blots, cells were lysed using 2x loading buffer. Lysate protein concentrations were measured using the Bio-Rad Protein Assay Dye Reagent (5000006). Extracted proteins were boiled at 100°C for 10 min, subjected to polyacrylamide gel electrophoresis using the Mini-PROTEAN system (Bio-Rad), and transferred onto nitrocellulose membranes using the Mini Trans-Blot Cell (Bio-Rad). Primary antibodies (listed in table S1) were suspended in 5% BSA in TBST and secondary antibodies were diluted in 5% milk in TBST. For histone blots, secondary antibodies were diluted in 5% BSA in TBST. Immobilon Western Chemiluminescent HRP substrate (EMD Millipore) or Clarity and Clarity Max ECL Western Blotting Substrates (Biorad, for histone blots) was used for visualization and imaging was performed using an ImageQuant 800 biomolecular imager (Amersham). For densitometric analysis, Western blot images were scanned using Adobe Photoshop (Adobe Systems Inc.) and quantified with the gel analysis macros available in FIJI (NIH, https://imagej.net/software/fiji).

Flow cytometry and FACS: Flow cytometry analysis was performed using a FACS LSRFortessa from BD Biosciences. Cell sorting was performed

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using a FACS Aria II SORP (4 or 5 lasers) from BD Biosciences. Analysis of flow cytometry and FACS data was performed using FCS Express software from De Novo Software. Cells for analysis were suspended in Annexin Binding Buffer with 50 ng/ml DAPI.

For initial experiments to determine the optimal transcription factor configuration, cells were plated at 150,000 cells per 6-well plate and transfected the next day (day 1). Medium changes were performed daily, starting on day 2 up to cell harvest and analysis on day 4.

For cells transduced with an αKG-RE promoter/GFP reporter and NtcA chimera in a single vector, cells were plated at 125,000 cells per well in a 6-well plate followed by three days of media changes [glutamine deprivation as well as dm-αKG (Sigma 349631) supplementation] before analysis.

For cells transduced with an αKG-RE promoter/GFP reporter and NtcA chimera in separate vectors, cells were plated at 250,000 cells per well in a 6-well plate and subjected to two days of media changes before analysis.

Cell cycle synchronization: To synchronize U251 cells at the G_1/S phase, cells were cultured with 2 mM thymidine (Sigma T1895) for 18 hours, washed with PBS, and then cultured in fresh media for 9 hours before incubating again with thymidine for 16 hours. To synchronize cells at the G2/M phase, cells arrested by thymidine treatment were cultured in fresh medium for 10 hours. Cells were stained with 20 μM Hoechst 33342 (Thermo Scientific 62249).

Immunofluorescence: 500,000 HEK293 or U251 cells were plated on 6 Micro-well glass bottom plates with 20 mm microwell 1.5 cover glass (Cellvis P06-20-1.5-N). The plate was pre-treated with 0.01% poly-L-ornithine solution (Sigma-Aldrich P4957) for HEK293 cells. The next day, 100 nM MitoTracker Red CMXRos (Cell Signaling Technology 9082S) was added in 2 ml cell culture medium and incubated at 37°C for 20 min. 4% paraformaldehyde was added to the cells after removing media with MitoTracker Red, followed by 0.2% Triton-X 1000 incubation and blocking with 0.2% fish skin gelatin (Sigma-Aldrich G7041). Primary antibody was added in 0.2% fish skin gelatin and incubated overnight at 4°C, including anti-FLAG Tag (1:800) or anti-HA Tag (1:1000) antibodies (listed in table S1). A 2 mg/ml stock of Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody conjugated to Alexa Fluor 488 was diluted 1:2000 in 1 ml of 0.2% fish skin gelatin and added after primary antibody and incubated at 4°C on a rocker for 2 hours. 300 nM DAPI in 0.2% fish skin gelatin was added for the last 5 min of incubation. Images were obtained in dual camera mode on a CSU-W1 spinning disk confocal microscope (Nikon).

Pyruvate quantification in blood and CSF: Absolute pyruvate concentrations in blood or CSF samples from children or adults were obtained from Geigy Scientific Tables, 1981 (*89*).

Liquid chromatography–mass spectrometry and stable isotope tracing: Cells were plated at a density of 250,000 cells per well in a 6-well plate for all LC-MS analyses unless otherwise stated. For relative quantification of metabolite abundance, cells were subjected to two days of media changes with supplementation of (*R*)-2HG-TFMB [a kind gift of R. Looper (University of Utah)], dimethyl succinate (Sigma-Aldrich W239607), or DMSO before collection and analysis.

For stable isotope tracing experiments, cells were plated at a density of 25,000 cells per well in a 6-well plate. Cells were cultured in the presence or absence of 1 mM pyruvate for three days, with media changes each day. Two hours before harvest, media was changed again to glutamine-free DMEM supplemented with 4 mM unlabeled glutamine or 4 mM α-^{15}N-glutamine, and with or without 1 mM pyruvate.

Samples were washed twice with ice-cold saline and snap frozen in liquid nitrogen. Cells were harvested by scraping and metabolites were extracted directly in 80% acetonitrile (1 μl per 1,000 cells) and vortexed for 20 min at 4°C. Cells were then centrifuged at maximum speed for 10 min in a microcentrifuge at 4°C. The supernatant was harvested and centrifuged an additional time to remove debris.

To quantify αKG from cell extracts using isotope dilution mass spectrometry, metabolites were extracted from cells as described above. 1,2,3,4-$^{13}C_4$, αKG (Cambridge Isotope Laboratories, CLM-4442-PK) was added to each extract so that each sample contained a final concentration of 200 nM internal standard. Alongside these extracts a calibration curve consisting of 10 calibrator points ranging from 10 nM to 10 μM αKG (Sigma-Aldrich K1128) was prepared. 1,2,3,4-$^{13}C_4$ αKG was added to each calibrator, so the final concentration of the internal standard was also 200 nM.

For absolute quantification of αKG, peaks for the native metabolite and internal standard were integrated in TraceFinder 5.2 SP1 software (Thermo Fisher). The known concentrations of each calibrator were added to the software to generate a linear regression curve with $1/X^2$ weighting. αKG levels were quantified against this curve. For each experimental condition, a duplicate cell sample was used to measure cell number and volume. The contents of each well were trypsinized and the cell number and volume were measured using a Beckman VI-CELL-XR Counter. Whole-cell αKG concentrations were calculated using the total moles in a whole-cell sample, the total number of cells per sample, and the volume of each cell. Cellular αKG concentration was calculated as $C_{\text{cell}} = \frac{C_{\text{meas}} V_{\text{ext}}}{N_{\text{cell}} \cdot \frac{4}{3}\pi\left(\frac{d}{2}\right)^3}$, where C_{cell} is the estimated intracellular αKG concentration, C_{meas} is the measured αKG concentration in the extract based on standard curve, V_{ext} is the final extract volume, N_{cell} is the total number of cells extracted, and d is the mean cell diameter. Cells were approximated as spheres, so the single-cell volume calculation is $\frac{4}{3}\pi(d/2)^3$.

Metabolite separation and concentration was achieved with a Millipore ZIC-pHILIC column using a linear gradient of 10 mM ammonium acetate in water, pH 9.8 (Mobile Phase A) and acetonitrile (Mobile Phase B), as previously described (*90*). Metabolites were detected with a Q Exactive HF-X or Orbitrap Exploris 480 (Thermo Scientific) at a resolving power of either 120,000 or 240,000 at full width at half maximum (FWHM), a scan range set to 80–1,200 m/z, and polarity switching, as previously described (*91*). To determine relative abundances of αKG, peaks were integrated using El-Maven 0.12.0 software (Elucidata) and total ion counts were quantified using TraceFinder 5.2 SP1 software (Thermo Fisher). Peaks were normalized to total ion counts using the R statistical programming language.

For stable isotope tracing studies, natural isotope abundance correction was performed using the AccuCor package in the R statistical programming language (*92*).

Gas chromatography–mass spectrometry: Quantification and analysis of steady state αKG levels were performed using methods previously described (*93*). Briefly, cells were plated in 6-well plates at a density of 125,000 per well. Media was changed daily for 3 days, and on day 4, cells were washed with ice-cold normal saline, and plates were snap frozen with liquid nitrogen. Metabolites were extracted by adding 350 μl ice-cold 70% methanol to each well. Cells were scraped off each well, and the cell suspension was transferred into an Eppendorf tube on dry ice. To each sample, 150 μl chloroform was added, and the samples were vortexed at 4°C for 20 min, followed by centrifugation at 17,000 g for 10 min. The upper layer of methanol containing polar metabolites was transferred to a separate tube and dried overnight on a vacuum rotary evaporator (CentriVap, Labconco) at 4°C. Samples not used immediately for analysis were stored at -80°C.

Dried samples were derivatized by adding 20 μl methoxamine (MOX, Thermo Fisher 45950) per sample and vortexed for 20 min at 4°C, followed by incubation at 37°C for 1 hour. 30 μl N-tert-Butyldimethylsilyl-N-methyltrifluoroacetamide with 1% tert-Butyldimethylchlorosilane (TBDMS, Sigma 375934) was then added to each sample. The samples

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were vortexed briefly at room temperature and incubated for 1.5 hours at 65°C.

GC-MS analysis was performed on derivatized samples using an Agilent 7890B GC/5977A MSD system. Peak integration was performed using the Metran software tool (*94*). Relative metabolite quantification analyses were performed by normalizing metabolite ion counts to total ion counts measured within each sample.

DNA methylation: For analysis of U251 stable lines, cells were cultured in DMEM supplemented with 10% FBS and 1% penicillin-streptomycin as described in the cell culture methods. After gene knockout, cells were divided into two groups: one supplemented with 200 μM ascorbate-2-phosphate and the other without. The cells were maintained under these conditions for 2.5 weeks before harvesting.

Genomic DNA (gDNA) was extracted from cultured cells or brain or kidney tissues of Gpt2 WT or Gpt2 KO mice using the Qiagen Blood & Tissue DNeasy kit, with RNase (NEB) added to remove RNA. The DNA was eluted in nuclease-free water and incubated with DNA Degradase Plus and Benzonase for four hours. Enzymatic reactions were quenched with methanol, and the samples were speed-dried. Finally, the dried samples were reconstituted in ammonium formate for UHPLC-MS/MS analysis. Samples were analyzed at the UT Southwestern Metabolic Phenotyping Core. 5 μl of sample was injected on Nexera X2 UHPLC coupled to an LCMS-8060 (Shimadzu Scientific Instruments) triple quadrupole mass spectrometer using the electrospray ion source in positive mode. 5-methyldeoxycytidine (5mC), 5-hydroxymethyldeoxycytidine (5hmC), and deoxycytidine (C) were analyzed by selective reaction monitoring using the following transitions: 5mdC 242 → 126, 5hmdC 258 → 124 and 258 → 142, dC 228 → 112. Targeted nucleosides were resolved on a Shimadzu Cell Culture Profiling Column using a gradient of solvent B MeOH/MeCN (1:1, v/v) 0.1% formic acid over solvent A H2O 0.1% formic acid at a 0.350 ml/min flow rate.

Detection of locus-specific 5mC levels was performed via duet multiomics solution evoC (biomodal). The duet multiomics solution evoC method (*95*) and analysis were performed according to the manufacturer's instructions, using 80 ng of genomic DNA as starting material from Gpt2 WT or Gpt2 KO mouse brain tissues.

Seahorse assays of mitochondrial function: Oxygen consumption rate (OCR) was assessed using the Seahorse XF Cell Mito Stress Test Kit. 10,000 cells in each well were plated one day before the assay. Next day, the culture medium was replaced with OXPHOS assay medium, which consisted of DMEM without phenol red, supplemented with 2 mM glutamine, 1 mM sodium pyruvate, and 10 mM glucose, adjusted to pH 7.4. The plate was pre-incubated at 37°C for 1 hour in a non-CO_2 incubator. OCR measurements were initially taken under basal conditions, followed by sequential injections of specific reagents: 2 μM oligomycin, an inhibitor of Complex V that enables the calculation of mitochondrial ATP production; 1 μM carbonyl cyanide-p-trifluoromethoxyphenylhydrazone (CCCP), an uncoupling agent used to determine maximal respiration and spare capacity; and finally, 1 μM antimycin A, inhibitor of Complex III to halt mitochondrial respiration and allow the determination of non-mitochondrial respiration.

Cell proliferation: Cell proliferation assays were performed using a Celigo Image Cytometer (Revvity). Cells were plated in 96-well plates at a seeding density of 5,000 cells per well. To construct growth curves, cells were incubated with 20 μM Hoechst 33342 (Thermo Scientific 62249) and 1.0 μg/ml propidium iodide (PI) (Thermo Scientific P3566) dyes and viable cell numbers were measured each day from day 0 to day 4. To ensure robustness, we performed three independent replicates, each in technical triplicate, for each condition.

Splice isoform analysis: Paired FASTQ RNA-seq files were uploaded into UT Southwestern Medical Center's high performance computing system (BioHPC) on an established Nextflow (v 20.01.0) platform for RNA-seq analysis called Astrocyte (v 2.1.0). Reads were trimmed using TrimGalore (v 0.4.1) and then aligned using HiSAT (v2.0.1). Picard (v 1.127) was then used to mark and remove duplicate reads. Transcript-level splice isoform expression analysis was performed using the HiSAT, StringTie (v 1.1.2), and Ballgown protocol which has been described (*96*). Ballgown is available through the Bioconductor suite through the R statistical environment (R v 4.1.1.). U251 RNA-seq data were obtained from the European Nucleotide Archive project (accession PRJEB3371) as well as from NCBI BioProject database (accession PRJNA631805) (*97*). Normal human astrocyte control RNA-seq data were obtained from NCBI (accession PRJNA631805) (*98*).

RNA-seq: Total RNA from 10 mg brain tissue samples was extracted using the RNeasy Mini Kit according to manufacturer's protocol (Qiagen 74004) and resuspended in 30 μl nuclease-free water. RNA-seq was performed and analyzed by Plasmidsaurus.

RNA-seq analysis: Gene ontology (GO) analysis was performed via the gene ontology platform (https://geneontology.org). Volcano plots were generated using ggplot2, displaying log-transformed *P* values on the *y* axis and log2 fold changes on the *x* axis. Gene set enrichment analysis (GSEA) was conducted using GSEA software (v.4.3.3) (https://www.gsea-msigdb.org). The normalized enrichment score (NES) and the false discovery rate (FDR) Q value were calculated by permuting gene set types, with a significance threshold of FDR ≤ 0.25 used to identify significantly enriched gene sets.

ChIP-seq: About 20 mg brain tissue for each sample and histone marker were lysed with a TissueLyser (Qiagen), then cross-linked with 1% formaldehyde in PBS for 10 min at room temperature. Cross-linking reactions were quenched with glycine for 5 min, followed by washes in cold PBS. The crosslinked tissue was homogenized using a cooled blade homogenizer in homogenization buffer (50 mM Tris-HCl pH 7.5, 1% Nonidet P-40, 0.25% deoxycholic acid, 1 mM EDTA). Then, cells were lysed with Farnham Buffer (5 mM PIPES, pH 8.0, 85 mM KCl, 0.5% Nonidet P-40, 1 mM DTT, 0.1 mM PMSF), followed by SDS Lysis Buffer (50 mM Tris-HCl pH 7.9, 10 mM EDTA, 1% SDS, 1 mM DTT) to obtain soluble chromatin.

Chromatin was sheared using a Covaris M220 Focused-ultrasonicator to generate DNA fragments approximately 200 to 400 bp in size, as verified by gel electrophoresis. The sonicated chromatin was centrifuged at maximum speed in a microcentrifuge for 1 min at 4°C to remove insoluble debris. The soluble chromatin supernatant was diluted 1:9 with Dilution Buffer (20 mM Tris-HCl pH 7.9, 300 mM NaCl, 2 mM EDTA, 0.5% Triton X-100, 1 mM DTT, 0.2 mM PMSF) supplemented with a protease inhibitor cocktail (Roche). The diluted chromatin was precleared with 20 μl/ml protein A-agarose beads (ThermoFisher Scientific) on a nutator for 1 hour at 4°C. After centrifugation at 1000 g for 3 min at 4°C, the supernatant was reserved for immunoprecipitation and 24 ng of spike-in chromatin (Active Motif 53083) was added per sample, with 2% set aside as input DNA. For immunoprecipitation, antibodies used included 5 μl H3K9me3 (Abcam AB8898 1063771-1), 5 μl H3K4me3 (Active Motif 39159) or 15 μl H3K27me3 (Cell Signaling 9733), along with 2 μg of spike-in antibody (Active Motif 61686), which were bound to 50 μl protein G Dynabeads (Invitrogen). The mixture was incubated with precleared supernatant overnight at 4°C with rotation. The Dynabeads were washed sequentially with the following buffers: low-salt wash buffer (10 mM Tris-HCl, pH 8, 2 mM EDTA, 0.1% SDS, 1% Triton X-100, 150 mM NaCl), high-salt wash buffer (10 mM Tris-HCl, pH 8, 2 mM EDTA, 0.1% SDS, 1% Triton X-100, 500 mM NaCl), LiCl wash buffer (10 mM Tris-HCl, pH 8, 1 mM EDTA, 1% NP-40, 1% Na-Deoxycholate, 250 mM LiCl), and a final wash with TE buffer containing 50 mM NaCl. Chromatin was eluted, incubated overnight at 65°C, and treated with

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RNase A and proteinase K. DNA was purified using the QIAquick PCR Purification Kit (Qiagen).

For library preparation, 3 ng of H3K4me3 ChIP DNA, 10 ng of H3K9me3 ChIP DNA, and 10 ng of H3K27me3 ChIP DNA were used with the NEBNext Ultra II DNA Library Preparation Kit for Illumina (NEB). Library quality was assessed using a High Sensitivity D5000 ScreenTape on an Agilent 2200 TapeStation and quantified with the Qubit dsDNA HS Assay Kit (Thermo Fisher). Libraries with unique adaptor barcodes were multiplexed and sequenced on an Illumina NextSeq 2000 platform (paired-end, 100 base pair reads), with a sequencing depth of 30 million reads per sample for H3K4me3 ChIP, 50 million reads for H3K9me3 ChIP, and 40 million reads per sample for H3K27me3 ChIP.

ChIP-seq analysis: Fastq files were trimmed with TrimGalore (v.0.6.10), and quality control was performed both before and after trimming using FastQC (v.0.12.1). ChIP-seq reads were aligned to the mm10 reference genome using Bowtie2 (v.2.5.1). The sam files were converted to bam files by Samtools (v.1.6). PCR duplicates were removed with PicardTools (v.3.0). Signals were normalized using the spike in values of the samples and the input. We chose Macs2 (v.2.2.9.1) to call the peaks using the broad peaks setting for H3K9me3 and H3K27me3 (FDR < 0.1); we used the narrow peaks for H3K4me3 (FDR < 0.1) in the analysis. ChIP-seq peaks that were significantly increased or decreased in the Gpt2 KO mice were then identified using the DiffBind (v.3.16.0) package in R (v.4.2). Bigwig files for heatmap generation were created from bam files using the bamCoverage function in deepTools. For visualization, representative track diagrams were generated using the Integrated Genomics Viewer (IGV) (v.2.9.4).

Subcutaneous xenotransplantation: Subcutaneous HCT116 xenografts were established by injecting HCT116 cells into the flanks of 5-week-old female NCR/NU mice (Taconic). Briefly, 10^6 cells were resuspended in a 100 µl suspension prepared by mixing 50 µl PBS with 50 µl Matrigel (Corning 354234) and injected subcutaneously into the flank. Four days after implantation, mice were treated with vehicle or 600 mg kg^{-1} dm-αKG by daily intraperitoneal injection. Tumor size was measured by calipers one week after implantation. Mice were sacrificed when tumors reached ~500 mm^3, and tumors were harvested for downstream analyses.

Statistical analysis: Statistical analysis was performed using Graphpad Prism software. Figure legends include information on all statistical tests performed. *P* values were calculated by unpaired *t* tests for tests of statistical significance involving comparison of two independent groups. For comparisons between an experimental condition and the designated control condition in immunoblot analyses of histone methylation marks in cultured cells, one-sample *t* tests were used to calculate *P* values in the experimental condition relative to a reference value established by the control condition (i.e., 1.0). For comparisons between two groups that did not include the designated control condition in immunoblot analyses of histone methylation marks in cultured cells, *P* values were calculated using paired *t* tests. For immunoblot analyses of histone methylation marks in tissues, *P* values were calculated using unpaired *t* tests. For tests of statistical significance involving comparison of three or more groups, *P* values were calculated by one-way ANOVA tests. For analyses of H3K4me3, H3K9me3, and H3K27me3 distributions in Fig. 6, A to C, *P* values were calculated by Kolmogorov-Smirnov tests. For the genome-wide 5mC analysis in Fig. 5D, *P* value was calculated by Wilcoxon signed rank test. For the comparison of gene expression changes in Fig. 5K, *P* value was calculated by χ^2 test. For comparisons of two groups with significantly different variances, Welch's *t* test was used. For all tests, *P* values less than 0.05 were considered statistically significant.

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S.K.M. and K.G.A. have intellectual property interests related to brain tumor metabolism (US patent application serial no. 63/597,523) and are cofounders of Gliomet. S.K.M. is a founder of Gliomic. R.J.D. is a founder and advisor at Atavistik Bioscience and an advisor for Vida Ventures, Illumina, and Faeth Therapeutics. **Data, code, and materials availability:** Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, S.K.M. (samuel.mcbrayer@utsouthwestern.edu). DNA constructs generated in this study are available upon request or from Addgene. Data are available in the main text or the supplementary materials. RNA-seq data have been deposited on Gene Expression Omnibus (GEO) (GSE325502). ChIP-seq data have been deposited on GEO (GSE325503). Synthetic DNA sequences have been deposited on GenBank. **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

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A transcriptional biosensor reveals mechanisms of #-ketoglutarate signaling to chromatin

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Editor's summary

Alpha-ketoglutarate (#KG) is a metabolite used by enzymes to chemically modify chromatin, which ensures appropriate gene regulation. However, how #KG is supplied to these enzymes in the nucleus is unclear. Sternisha *et al*. transferred an #KG-sensing transcription factor from bacteria to human cells to monitor nuclear #KG. They found that #KG produced by glutamate pyruvate transaminase 2 in mitochondria is a key source of #KG that is shuttled to nuclei. Loss-of-function mutations in this enzyme lead to dysregulated chromatin structure and misexpression of genes that are required for brain development. —Stella M. Hurtley

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